University of Moratuwa

Faculty of Engineering

Department of Electronic & Telecommunication Engineering

EN2160 - Electronic Design Realization

Restaurant Robot

Design Report

Team Members:

1. Arachchi M.B.F 220040T
2. Bandara I.W.T.N 220061H
3. Boralugoda M.S 220074B
4. Karunathilake K.D.K 220310X
5. Peiris P.I.U 220452H
6. Sanjeewa P.M.G.P.N 220577U
7. Senaweera S.A.H.D 220596C
8. Viduranga J.K.A 220663F
9. Wijayarathna K.K.B.C 220705M
10. Wijenayaka M.B.T.I 220711D

July 11, 2025

Contents

1 Introduction

1.1 Background of the project

Due to swift progress in service robotics and the growing need for automation in the hospitality sector, restaurant service robots have surfaced as a successful way to improve operational efficiency and increase customer experience.The worldwide shift towards contactless services, particularly during the pandemic, has expedited the incorporation of robots in customer-oriented settings.Reference models such as the BellaBot have illustrated the viability of these robots in traversing challenging situations and executing delivery missions with considerable reliability.

Our project aims to create a restaurant robot specifically designed for indoor settings, like cafés and dining halls. It is engineered to autonomously deliver food and engage with consumers, thereby lessening the burden on human personnel and reducing service inaccuracies.This robot provides a practical and scalable solution for the future of smart eating experiences through the integration of efficient mechanical design, robust control systems, and user-centered interaction elements.

2 System Overview

2.1 Description

Drawing inspiration from existing commercial solutions like BellaBot, this system integrates mobility, interactive communication, and intelligent control within a compact, modular design.The robot is powered by the SAM3X8E microcontroller, which offers the necessary computational capabilities for controlling motors, processing sensor data, and managing display and communication tasks. The entire system is structured around multiple interconnected subsystems that communicate via a robust RS-485 full-duplex bus, allowing synchronized and reliable data exchange between modules.

2.2 User Need Analysis

2.2.1 User Needs and Expectations

1. **Efficient and Accurate Food Delivery -** Customers expect robots to deliver food quickly and accurately to the correct table.

2. **User-Friendly Interaction -** Robots should have an easy-to-use interface for customers and restaurant staff.

3. **Navigation and Obstacle Avoidance -** Robots must safely navigate through a crowded restaurant while avoiding obstacles.
4. **Food Pickup Automation -** Kitchen staff require the robot to stop at a designated area to load orders.
5. **Hygiene and Safety Compliance -** Ensuring proper sanitization measures and food safety standards.
6. **Order Status Updates -** Customers may want real-time updates on their orders via displays or audio prompts.

2.2.2 Stake Holder Mapping

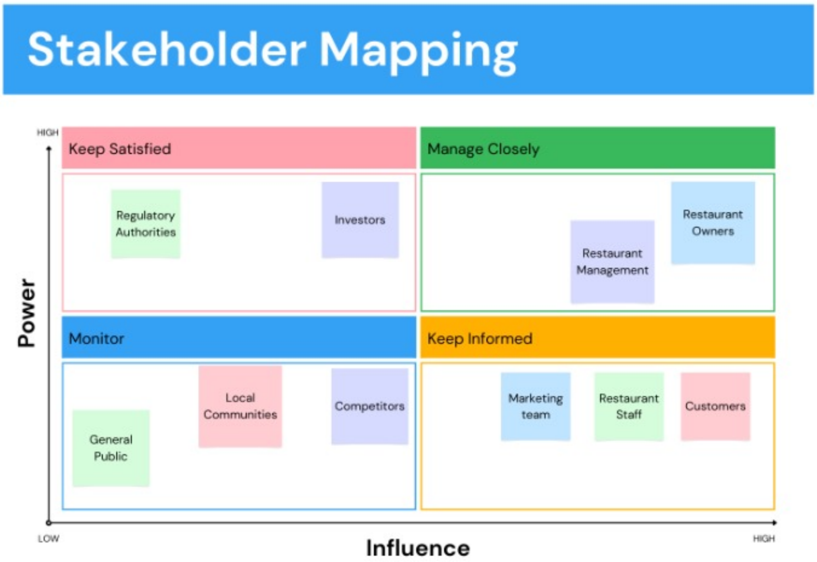


Figure 1: Stakeholder Map

2.3 Key Features in our Robot:

- **Autonomous Navigation:** The robot is equipped with inductive proximity sensors (OMRON E2E) and object avoidance modules (A02 series), allowing it to navigate safely and efficiently through dynamic environments such as crowded dining areas.
- **Interactive Display System:** A 7.0" TFT LCD with SSD1963 controller is used to display menus, guide customers, and confirm orders, ensuring a user-friendly interface.

- **Voice Feedback Capability:** The robot uses a speaker system powered by the MAX98357A DAC, Winbond W25Q128JV flash memory, and AS041008C speaker to provide audio feedback, improving human-robot interaction and approachability.
- **Tray Delivery Mechanism:** The robot supports multiple trays, each capable of holding up to 10 kg. The mechanical design focuses on balance and safety during motion.
- **Robust Communication Bus:** Half-duplex RS-232 communication is used between distributed modules, ensuring real time data transmission with high noise immunity and support for multiple nodes.
- **Efficient Power System:** A replaceable battery provides 12–24 hours of operation per charge, with a recharge time of approximately 4.5 hours, allowing for extended service during peak hours.
- **Structural Reliability:** Finite Element Analysis (FEA) using SolidWorks confirmed the design of the tray can handle forces up to 265 N with a safety factor of 2, ensuring mechanical robustness under load.
- **Safety Features:** The robot includes emergency stop functionality, collision detection, and is built with hygiene compliant materials suitable for food service environments.
- **Modular and Scalable Architecture:** The robot's design allows for future expansions and upgrades. Each subsystem is modular making maintenance easier and improving adaptability to different restaurant layouts or requirements.

3 Review of Reference Product

The design and functionality of the restaurant robot were inspired by **BellaBot**, a commercial service robot developed by **Pudu Robotics**. BellaBot is widely used in the hospitality industry for efficient and contactless food delivery.

Figure 2: Bella bot by Pudu Robotics

Specifications

- **Manufacturer:** Pudu Robotics
- **Height:** 1290 mm
- **Width:** 565 mm
- **Depth:** 537 mm
- **Weight:** Approximately 55 kg
- **Load Capacity:** Up to 40 kg (10 kg per tray, with 4 trays)
- **Battery Type:** Lithium-ion
- **Battery Life:** 12–24 hours on a full charge

- **Charging Time:** Approximately 4.5 hours
- **Navigation:** Laser SLAM + Visual SLAM dual system
- **Touch Interface:** 10.1" touchscreen display
- **Obstacle Avoidance:** Multi-sensor fusion for 360° environmental perception
- **Interactive Features:** LED cat-like facial expressions, voice prompts, ambient lighting effects
- **Speed:** Adjustable, up to approximately 1.2 m/s

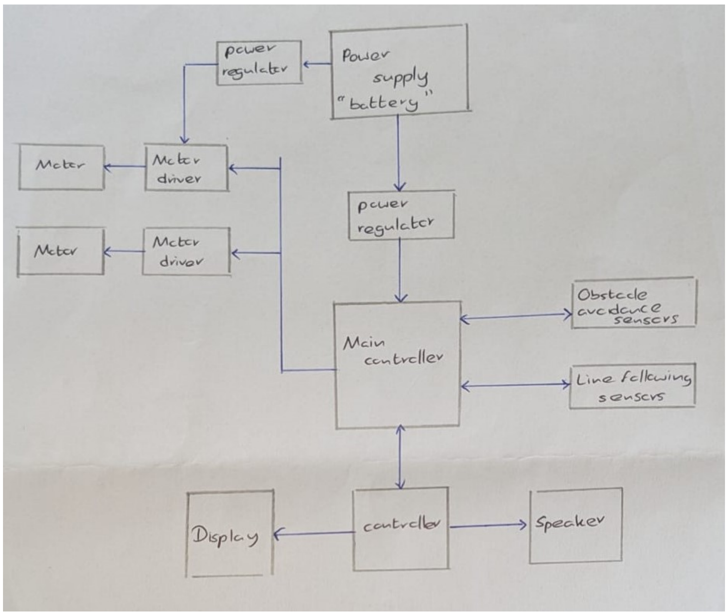
Key Functionalities

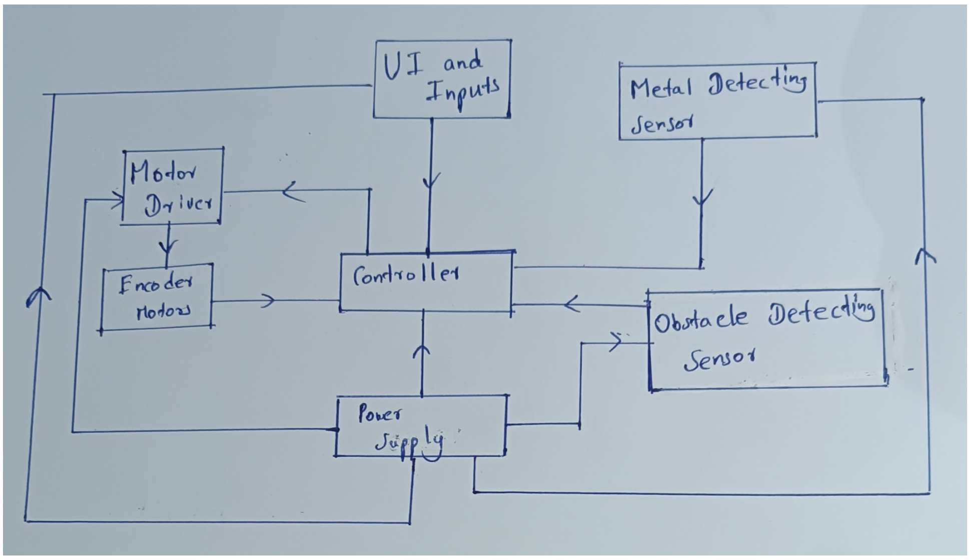
- Autonomous navigation with multi-point delivery
- Multi-modal interaction (touchscreen, voice, LED facial expressions)
- Automatic return to charging dock
- Obstacle avoidance in dynamic environments

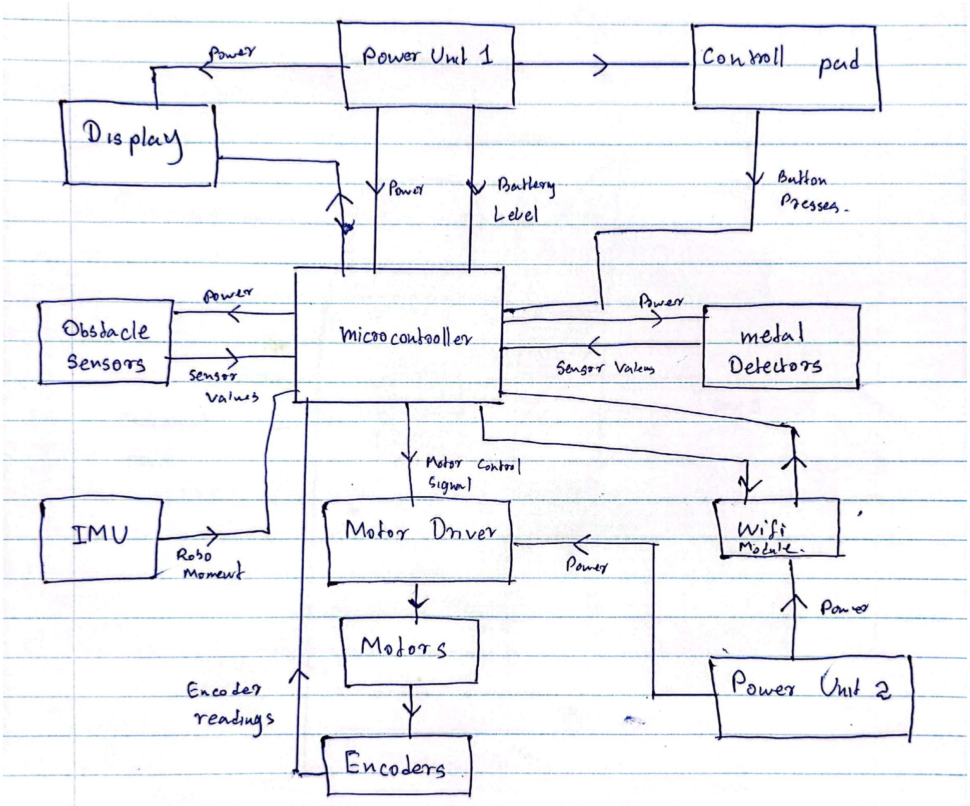
Note: The BellaBot served as the reference model during conceptualisation and design requirement analysis for this project to ensure practical feature selection, ergonomic considerations, and user experience alignment. While BellaBot uses Laser SLAM and Visual SLAM for navigation, our product adopts a metal line following method, which is more suitable and cost-effective for local restaurant settings.

4 Block diagram

4.1 Conceptual Design Stage





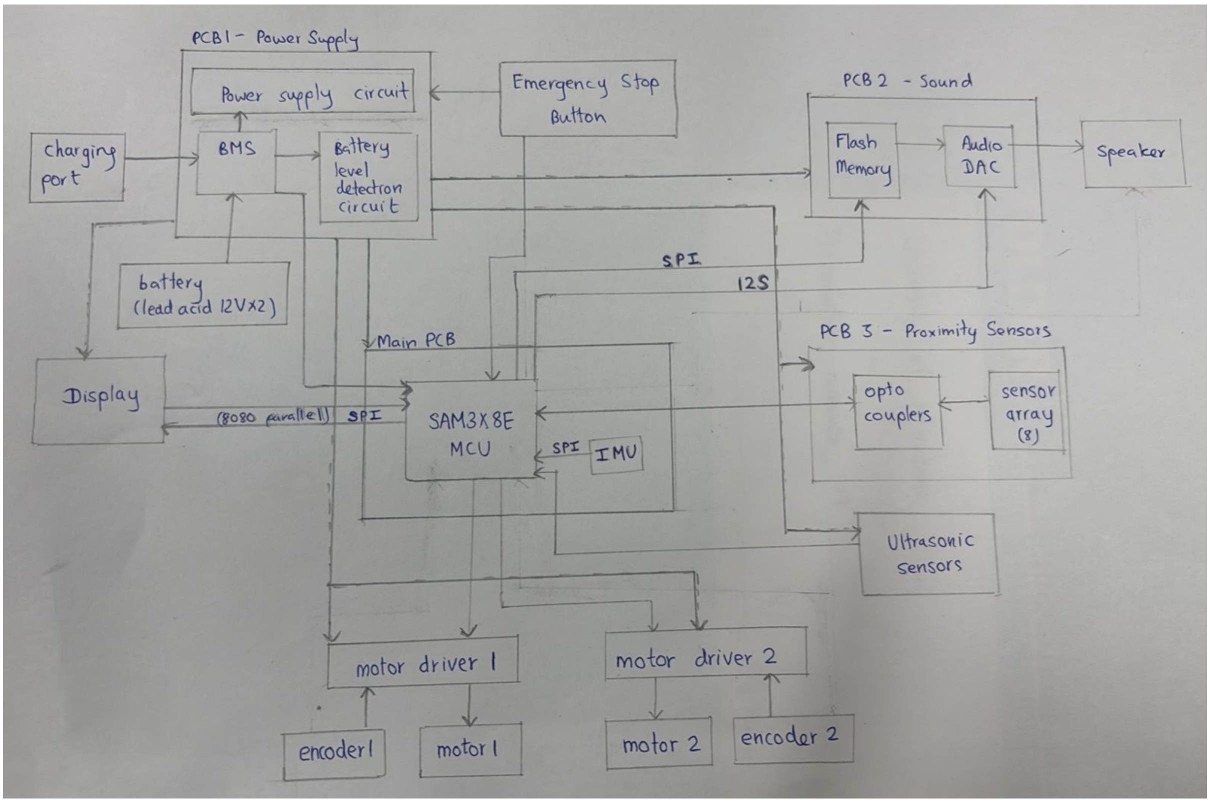


4.2 Evaluation

Table 2: Block Diagram Evaluation

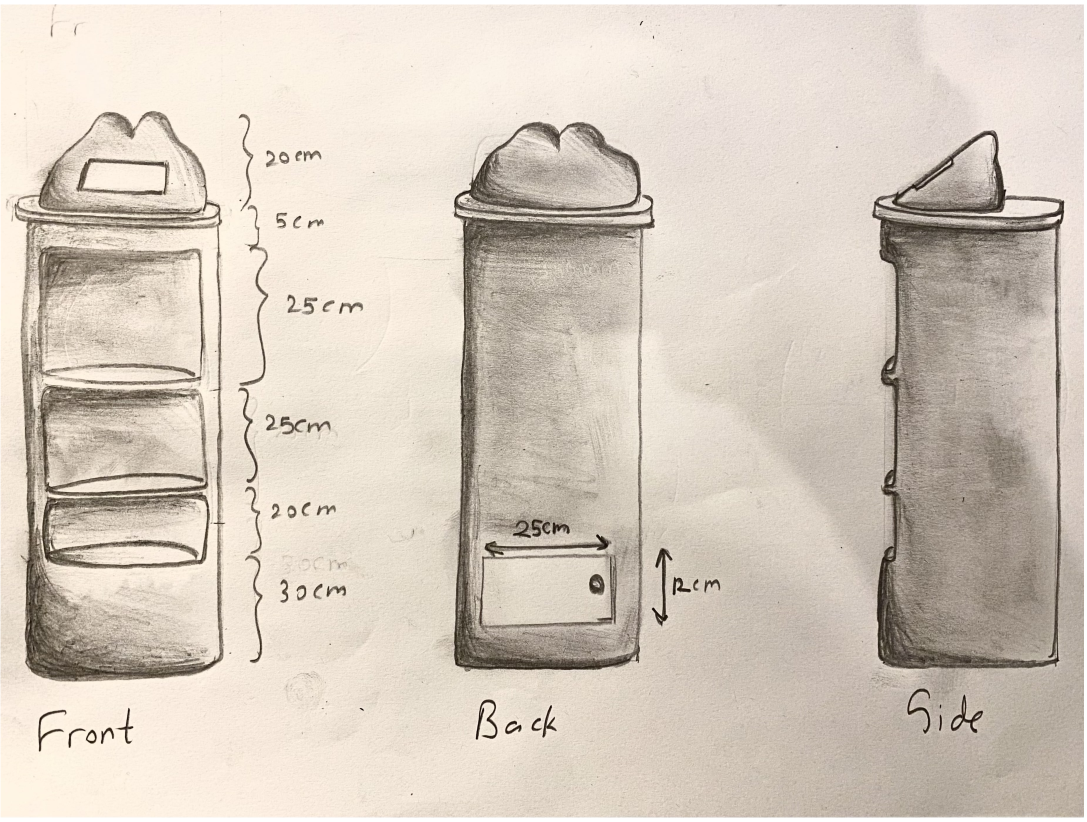
Criteria	220040T	220061H	220074B	220310X	220452H	220577U	220596C	220663F	220705M	220711D
Completeness of components	3	4	3	3	3	3	4	4	4	4
Fulfillment of main functionality	4	4	3	4	4	4	3	4	4	4
Accuracy of interconnections	3	3	3	3	3	3	4	3	3	3
Power management	4	4	3	3	3	3	3	4	3	4
Overall	14	15	12	13	14	13	14	16	14	15

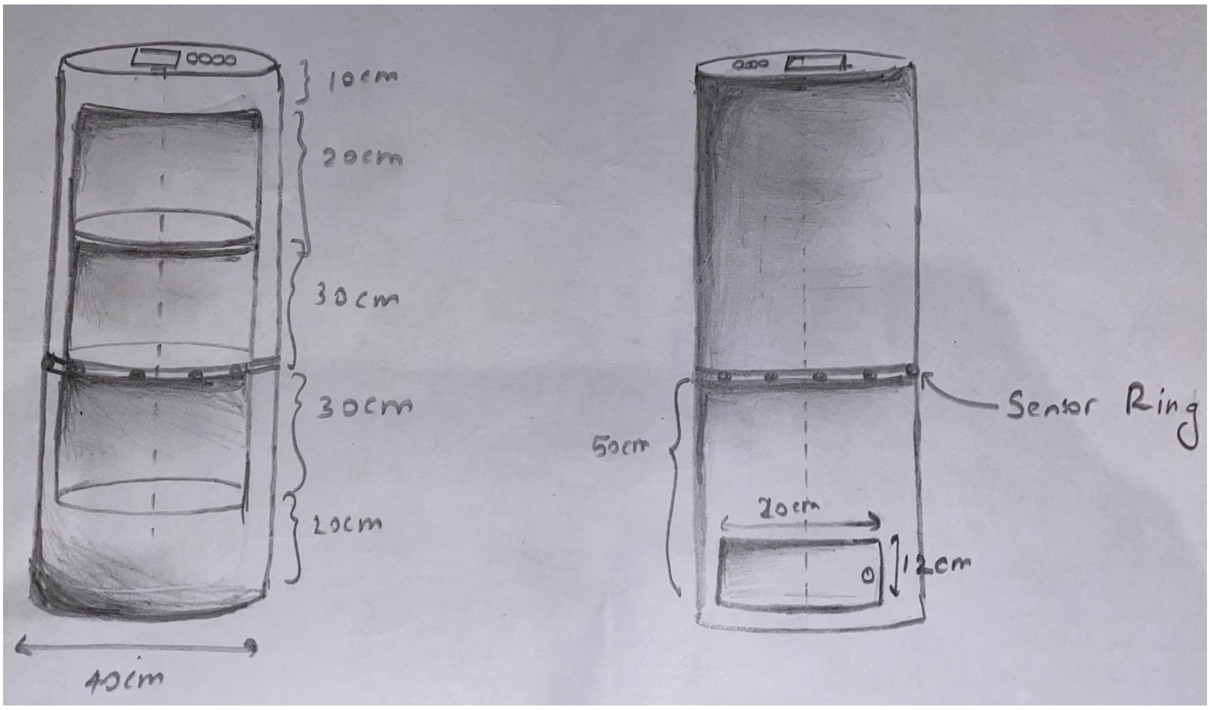
4.3 Final block diagram

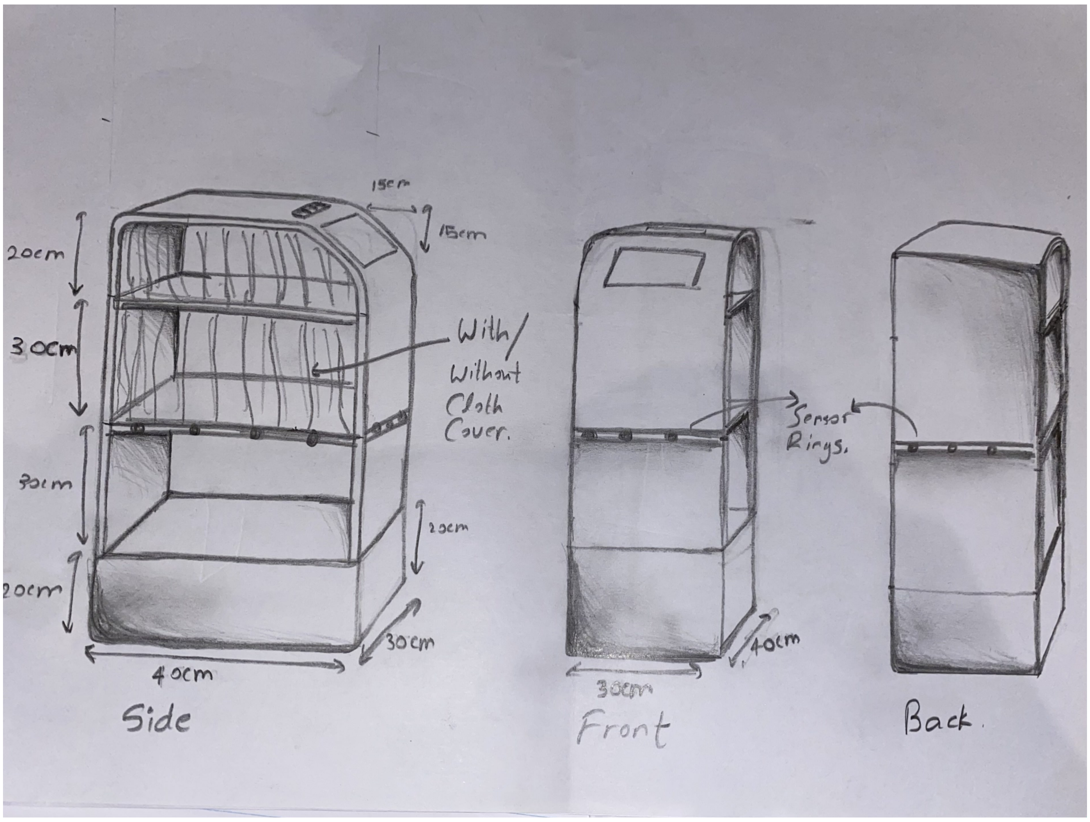


5 Mechanical Design

5.1 Conceptual Design Stage





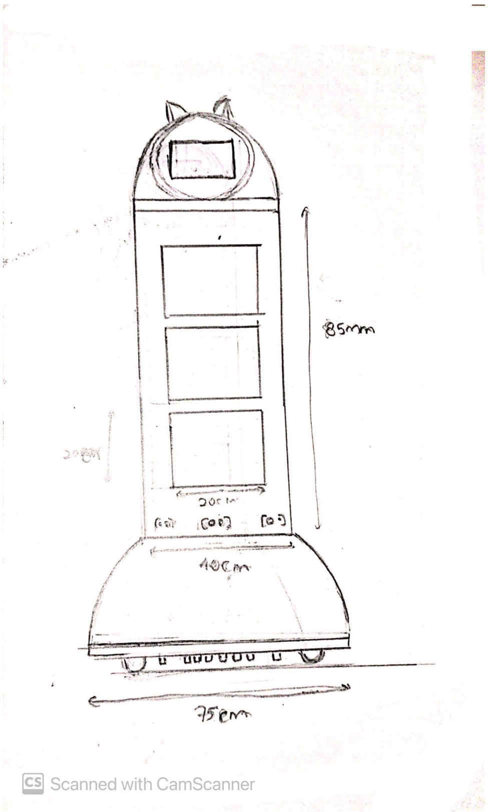


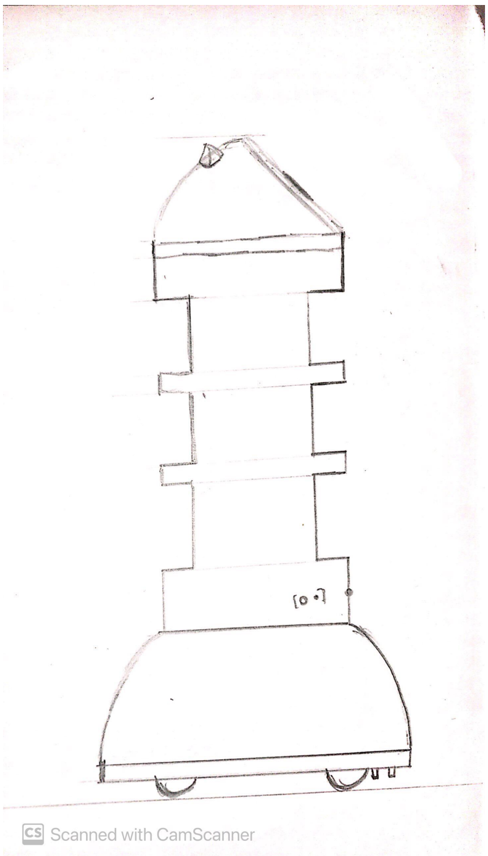
5.1.1 Evaluation

Table 1: Enclosure - Conceptual Design Evaluation

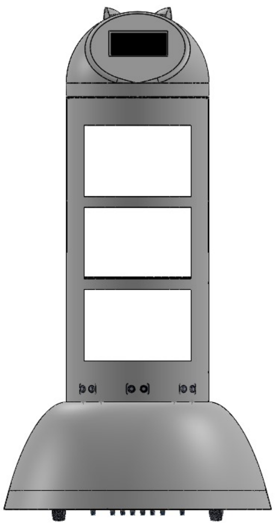
User Need	220040T	220061H	220074B	220310X	220452H	220577U	220596C	220663F	220705M	220711D
Efficiency	3	3	4	3	4	4	4	3	3	4
User friendly interaction	4	3	4	4	4	4	4	4	4	3
Customization and adaptability	2	3	3	5	2	3	4	1	4	2
Navigation and obstacle avoidance	3	4	3	4	3	4	4	3	4	3
Hygiene	4	3	3	4	2	3	3	4	3	2
Design	3	4	4	5	3	4	4	4	4	5
Overall	19	20	21	25	18	22	23	19	22	19

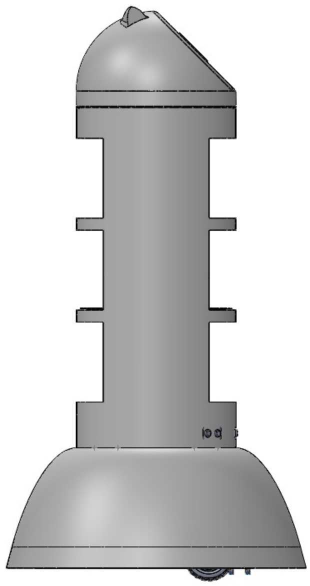
5.2 Final Cross Pollinated Design

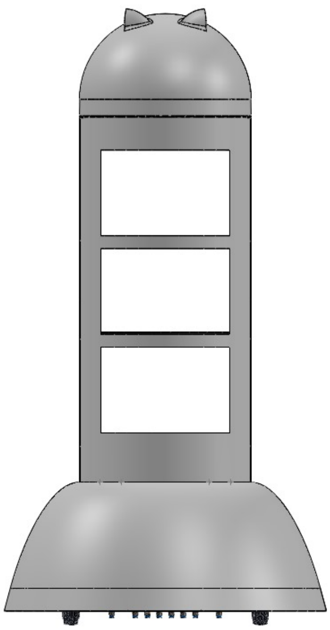




5.3 Solidworks Enclosure Design







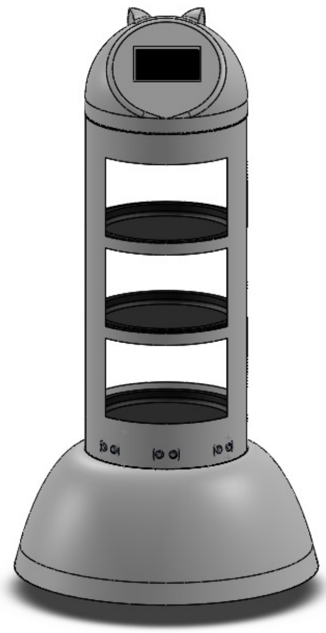


Figure 3: Frame designed using steel

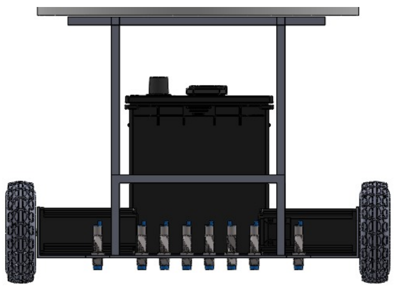
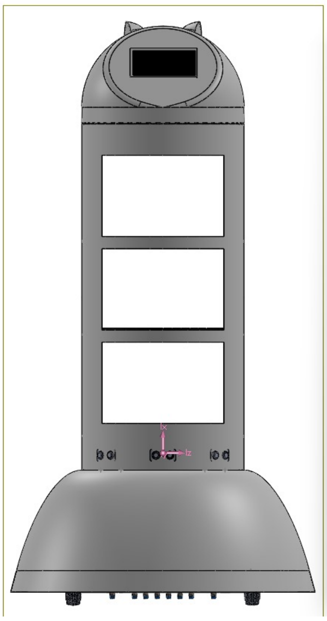
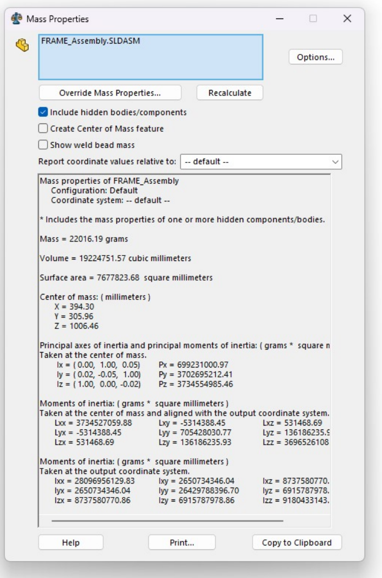
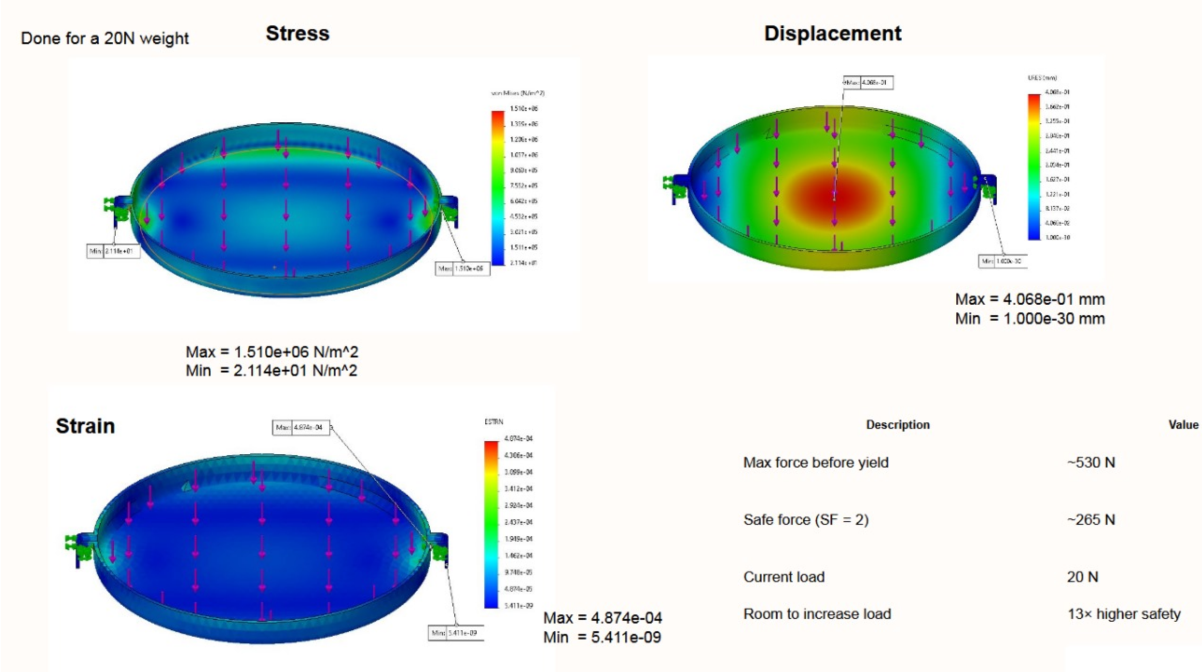


Figure 4: Plate designed using plastic

5.4 Analysis







Tray Weight Analysis and Stress Simulation

6 Hardware Design

6.1 Sensors

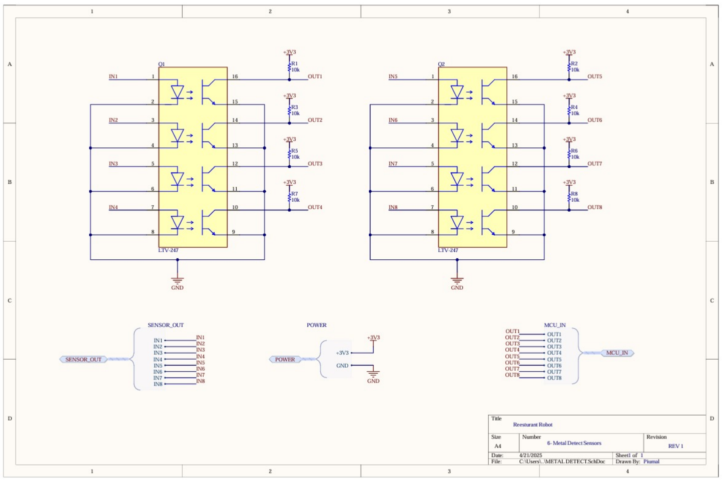
6.1.1 Metal Detecting Sensors

1. OMRON welding proximity sensors(E2EW Series)
2. OMRON welding proximity sensors(E2E/E2EQ Next Series)
3. IMB Inductive proximity Sensor
4. Balluf Inductive Proximity Sensors

Table 1: Comparison

Feature	OMRON E2EW	OMRON E2E/E2EQ	SICK IMB	Balluff BES004E
Sensitivity	3	3	2	3
Detection Range	2	3	3	3
Response Time	2	3	3	3
Robustness	3	2	3	3
Harsh Environment Suitability	3	2	3	3
Mounting Flexibility	2	3	3	3
Interface Options	2	3	2	2
Ease of Integration	2	3	2	2
Availability	3	3	2	3
Price Efficiency	2	3	2	2
Total Points	**24**	**28**	**25**	**27**

After carefully considering all the options available, we have chosen the **OMRON welding proximity sensors(E2E/E2EQ Next Series)** for our project.



6.1.2 Obstacle Detecting sensors

1. **A02 series sensor module**

Table 2: A02 series sensor module

Features	Advantages	Applications
3cm small blind zone	Small blind zone (5cm)	Horizontal distance sensing
3.3–5V power input	Water and dust resistivity	Car parking system
Closed waterproof bistatic probe	Strong anti-interference ability	Robot avoidance and automatic control

2. **ME007YS Series Sensor Module**

Table 3: ME007YS Sensor Module

Features	Advantages	Applications
Intelligent signal processing circuit, small blind zone	High-accuracy distance sensing	Horizontal distance sensing
Multiple interfaces: PWM, UART Auto, UART Ctrl, Switch	Stable temperature-compensated sensing	Solid level monitoring

3. **A13 Series Sensor Module**

Table 4: A13 Series Sensor Module

Features	Advantages	Applications
Reflective structure with narrow beam angle	High sensitivity and stable detection	Horizontal distance sensing
Multiple output modes: UART, PWM, RS485, Switch	Internal temperature compensation with anti-static design	Object classification and proximity sensing

Table 5: Ultrasonic Sensor Comparison

Feature	A02 Series	ME007YS Series	A13 Series
Blind Zone	3	1	2
Measuring Range	3	2	1
Water resistance	3	2	0
Dust resistance	3	2	0
Beam Angle	3	3	1
Static Power Consumption	3	2	2
Operating Current	3	3	3
Temperature Range	3	3	3
Accuracy	3	3	3
Response Time	3	2	2
Interface Variety	2	3	3
Form Factor / Size	3	2	2
Stability	3	2	3
Anti-Static Design	2	3	3
Temperature Compensation	2	3	3
Total Points	**42**	**36**	**31**

After carefully considering all the options available, we have chosen the **A02 Series sensor** for our project.

6.2 Actuators

6.2.1 Motors

1. **Calculations**

   (a) **Robot Requirements**

   - **Total robot mass:** 45 kg
   - **Drive system:** 2 direct-drive wheels (differential configuration)
   - **Maximum speed:** 0.5 m/s
   - **Acceleration:** 0.2 m/s^2
   - **Wheel diameter:** 127 mm (5 in)
   - **Indoor use only:** Low noise, non-marking tread required

   (b) **Motor Torque Calculation**

   - **Force per Wheel :**

   $$F = \frac{45 \cdot 0.2}{2} = 4.5\,\mathrm{N}$$

   - **Rolling Friction per Wheel:** For rubber wheels on carpet,the rolling resistance coefficient(Crr) is assumed to be in the range 0.03–0.05.For this calculation,we use an estimated value of Crr=0.05.

   $$F_r = \frac{C_{rr} \cdot m \cdot g}{n} = \frac{0.05 \cdot 45 \cdot 9.81}{2} = \frac{22.0725}{2} = 11.036\,\mathrm{N}$$

   - **Torque Requirement :**

   $$\tau = F \cdot r = (4.5 + 11.04) \cdot 0.0635 = 0.98679\,\mathrm{N \cdot m}$$

   Apply 2× safety margin:

$$\tau_{\text{required}} \approx 1.9736\,\text{N}\cdot\text{m}$$

- **Required RPM:**

$$\text{Circumference} = 2\pi r = 0.399\,\text{m} \quad \Rightarrow \quad \text{RPM} = \frac{0.5}{0.399}\cdot 60 \approx 75\,\text{RPM}$$

2. **Comparison**

Table 6: Motor Comparison

Motor	Torque	Control	Efficiency	Cost	Integration	Feedback	Total
Pololu 37D	15	10	10	12	14	0	61
Cytron SPG50	14	10	10	14	14	0	62
NEMA 23 (Open)	22	15	9	12	13	0	71
NEMA 34 (Open)	25	15	8	10	10	0	68
NEMA 23 Closed-Loop	**24**	**20**	**13**	**13**	**14**	**10**	**94**
JMC Servo-Stepper	25	19	12	10	12	10	88

After carefully considering all the options available, we have chosen the **NEMA 23 Closed Loop Stepper Motor** for our project.We decided to purchase YS Series 1 Axis Closed Loop Stepper CNC Kit from stepper online which includes a **S Series Nema 23 Closed Loop Stepper Motor** and a **CL57Y Y Series Closed Loop Stepper Driver.**

3. **Motor Specifications**

- **Manufacturer Part Number:** 23HS40-5004D-E1K-1M5
- **Motor Type:** Bipolar Stepper
- **Step Angle:** 1.8°

- **Holding Torque:** 3.00 Nm (424.83 oz.in)
- **Rated Current per Phase:** 5.0 A
- **Phase Resistance:** 0.7 Ω ± 10%
- **Inductance:** 2.3 mH ± 20% (1 kHz)
- **Insulation Class:** B (130°C / 266°F)
- **Frame Size:** 57 × 57 mm
- **Body Length:** 122 mm
- **Shaft Diameter:** 8 mm
- **Shaft Length:** 22 mm
- **D-Cut Length:** 15 mm
- **Cable Length:** 1500 mm

4. **Encoder Specifications**

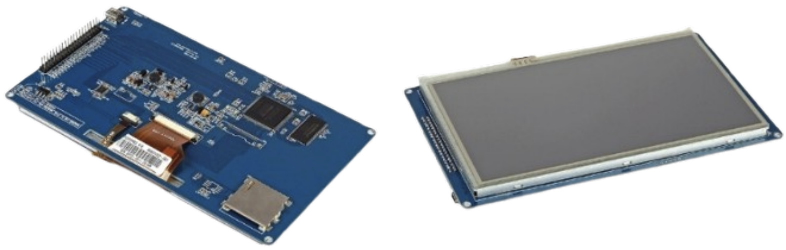
- **Output Circuit Type:** Differential type
- **Encoder Type:** Optical Incremental
- **Encoder Resolution:** 1000 PPR (4000 CPR)
- **Output Signal Channels:** 2 channels
- **Supply Voltage Range:** 4 V – 5 V
- **Output Current:** 100 mA
- **Output High Voltage:** 5 V
- **Output Low Voltage:** 4.5 V
- **Maximum Output Frequency:** 60 kHz

5. **Driver Specifications**

- **Output Peak Current:** 0–7 A (0–5 A RMS)
- **Input Voltage:** +24–50 VDC (Typical: 36 VDC)
- **Logic Signal Current:** 7–16 mA (Typical: 10 mA)
- **Pulse Input Frequency:** 0–200 kHz
- **Pulse Width:** 2.5 μs
- **Isolation Resistance:** 500 MΩ

6.3 Display and UI

The restaurant robot project aims to enhance customer experience by automating tasks such as taking orders, displaying menus, and providing real-time updates. A display module is critical for this purpose, serving as the primary interface for user interaction and information sharing and we will be using a **7.0" TFT LCD Module with SSD1963 Controller** for our application.



6.3.1 Display Specifications

Parameter	**Specification**
Model	7.0" TFT LCD Module
Resolution	800×480 pixels
LCD Type	TFT Transmissive Normal White
Interface	16-bit parallel bus interface
PCB Color	Blue
Backlight Power Supply	On-board 400 mA DC-DC Boost regulator
Module Dimensions	186 mm × 106 mm × 23 mm
Active Area	154 mm × 86 mm
Pixel Pitch	0.179 mm × 0.179 mm
Pin Header	Standard 2×20 2.54 mm pin header
Weight	410 grams

6.3.2 Electrical Specifications

Parameter	**Specification**
Driver Chip	SSD1963
Data Port Voltage	IOVCC = VCC = 2.8V 3.3V
Operating Temperature	-20°C to +70°C
Storage Temperature	-30°C to +80°C
Touch Type	Resistive touch
Connection Method	2.54mm pin header

6.3.3 Advantages

- High resolution (800×480 pixels) ensures crisp visuals.
- Resistive touch screen enables user interaction.
- Integrated backlight regulator simplifies power management.
- Pre-tested initialization code saves development time.
- Wide compatibility with the STM32 microcontroller.

6.4 Sound Module

The sound system in the restaurant robot is designed to provide a high-quality audio output for user interactions. The following components are selected for optimal performance.

6.4.1 Required Components

1. **I2S to Analog Converter Options**

   - **PCM5102A :** The PCM5102A is used as the I2S DAC to convert digital signals from the MCU into high-fidelity analog audio output. It supports up to 32-bit, 384 kHz audio and has a built-in low-noise voltage regulator for stable operation.
   - **MAX98357A (I2S DAC + Amplifier)** The MAX98357A is an integrated solution that combines an I2S DAC with a Class-D amplifier, providing both digital-to-analog conversion and speaker driving capabilities in a single package. This space-efficient solution simplifies the audio signal chain by eliminating the need for a separate amplifier IC.Key features of this are :
     - Integrated I2S audio DAC and Class-D amplifier

- 3.2W output into 4Ω speaker at 5V
- No external components required for filtering
- Low quiescent power consumption
- Filterless modulation scheme

2. **Speaker Driver IC Options**

   - **PAM8403 :** The PAM8403 Class-D amplifier is a smaller, lower power option suitable for applications where space and power consumption are critical. It provides up to 3W per channel and is easy to implement.
   - **LM386 :** The LM386 is a classic low-voltage audio power amplifier that can drive speakers directly. It offers adjustable gain and volume control capabilities through simple external components.Some of the key features are
     - Operating voltage: 4V–12V
     - Output power: 0.25W to 1W (depending on supply voltage and load)
     - Adjustable gain from 20 to 200
     - Low quiescent current drain
     - Hardware volume control capability via external potentiometer
     - Simple implementation with minimal external components

3. **Flash Memory :** For storing pre-recorded audio files, the W25Q256 (32MB) or W25Q128 (16MB) NOR Flash memory is used. It communicates with the MCU via SPI and ensures fast, non-volatile storage for voice prompts and sound effects.

4. **Speaker Options**

   - **AS07108PO-LW152-WR-R (PUI Audio) :**A compact yet powerful speaker is required to match the amplifier's output. A 3W-5W, 8Ω full-range speaker is chosen to deliver clear voice playback with a sensitivity of 86dBA.
   - **Visaton K 50 SQ (4ohms, 2W)**
     - **Power handling:** 2W RMS
     - **Impedance:** 4 ohms
     - **Sensitivity:** 84 dB (1W/1m)
     - **Type:** Square dynamic speaker, paper cone

6.4.2 Design Considerations

1. **Option 1: PCM5102A + PAM8403**
   - Highest audio quality
   - Digital volume control via STM32
   - Higher component count
   - More PCB space required

2. **Option 2: PCM5102A + LM386**
   - Good audio quality
   - Hardware volume adjustment possible
   - More analog components required
   - Lower power output (0.25W–1W)

3. **Option 3: MAX98357A only**
   - Simplest implementation
   - Smallest PCB footprint
   - Digital volume control only
   - Good power output (3.2W at 5V)

   MAX98357A provides the simplest implementation by integrating both the amplifier and digital volume control in a single chip, eliminating the need for additional analog components and reducing system complexity. Its small footprint makes it ideal for designs with tight space constraints.Despite its compact size, the MAX98357A delivers 3.2W of power output at 5V, which is sufficient for many audio applications, offering efficient amplification without the need for a separate power amplifier.

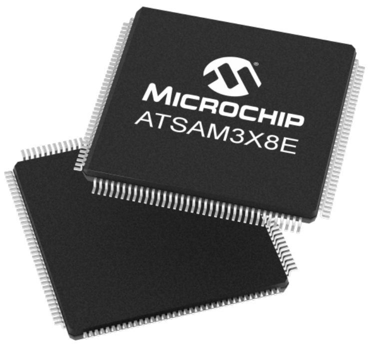
6.5 Microcontroller

6.5.1 Evaluation

After carefully considering all the options available, we have chosen the **SAM3X8E** microcontroller for our project.

Table 7: Microcontroller Comparison

Feature	MX RT1064	PIC32MZ	SAME70	SAM3X8E	Teensy 3.5/3.6	ATSAM4S	ATSAM3S
Core	3	2	3	2	3	3	2
I/O Pins	3	3	2	2	3	2	1
Memory	3	3	2	1	3	2	2
Speed	3	3	2	1	3	2	2
USB Support	3	3	2	2	3	2	2
Communication	3	3	3	2	3	3	2
Motor Control	3	2	2	2	3	3	2
DMA Support	3	2	3	0	2	2	2
Power Efficiency	2	2	2	3	2	3	3
Touch Interface	3	3	3	2	2	2	1
Parallel Interface	3	2	2	3	1	2	1
Development Tools	3	2	2	3	3	3	2
Community Support	1	2	3	2	3	2	2
Cost Efficiency	0	1	1	3	0	2	3
Ease of Programming and Debugging	2	2	2	2	3	2	2
Low Power Modes	2	1	2	2	2	3	3
Total Points	**41**	**38**	**41**	**42**	**40**	**42**	**39**



6.5.2 Details of SAM3X8E

1. **Core and Speed :** We chose the **SAM3X8E** because it features an ARM Cortex-M3 core running at 84 MHz. Although there are faster options available, this clock speed provides the perfect balance between performance and power consumption for our requirements. The SAM3X8E is powerful enough to handle tasks like driving the SSD1963 LCD, managing motor control, and processing sensor data, without being overkill for our project.

2. **I/O Pins :** We need a microcontroller with sufficient digital and analog I/O pins to handle the requirements of the display,motor controller, sensor inputs, and sound module. The SAM3X8E offers 54 digital I/O pins and 12 analog pins, which is more than enough to support

   - **SSD1963 :** The parallel interface (16-bit) is fully supported with available digital pins
   - **Motor Control:** We can easily control multiple motors using PWM signals.
   - **Sensor Inputs:** The 12 analog pins give us ample capacity to handle sensor data.

- **Sound Module:** We have enough I/O pins to interface with the sound module.

3. **Communication Interfaces :** For communication, we need a microcontroller that supports protocols like SPI, I2C, and UART. The SAM3X8E supports all these protocols, which gives us the flexibility to interface with the SSD1963 LCD, motor controllers,sensors, and sound modules. The microcontroller's ability to handle these interfaces makes it ideal for our design.

4. **Development Tools :** We also considered the availability of development tools, and the SAM3X8E comes with support for Atmel Studio, which is a free and powerful tool for development. It's a great advantage to be able to use a professional IDE without any additional costs.

5. **Parallel Interface (16-bit/8-bit) :** A key feature for us was the support for a 16-bit parallel interface. This is critical for driving the SSD1963 display, as parallel communication allows for faster data transfer compared to serial interfaces. The SAM3X8E provides this interface, ensuring that we can update the display quickly and efficiently.

6. **Cost :** We also found that the SAM3X8E is reasonably priced compared to other microcontrollers with similar features. It offers a good balance between performance and cost, making it an affordable option for our project.

6.5.3 Power Requirements Analysis

The table below shows the power consumption of all robot components:

Component	Quantity	Voltage (V)	Current (A)	Power (W)
Nema 23 motors	2	24	4.2	201.6
AO2 PWM Ultrasonic sensors	3	5	0.04	0.6
Proximity sensors	8	12	0.003	0.288
7-inch TFT LCD Display	1	5	0.12	0.6
SAM3X8E MCU	1	3.3	0.8	2.64
Motor drivers CL57T (logic)	2	24	0.016	0.768
Display backlight (21 LED)	1	12	0.14	1.68
Total Power Consumption				≈ 210

6.5.4 Battery Capacity Calculation

Short-Duration Operation (2 hours)

- Energy requirement: $210 \text{ W} \times 2 \text{ h} = 420 \text{ Wh}$
- Capacity at 24V: $\frac{420 \text{ Wh}}{24 \text{ V}} = 17.5 \text{ Ah}$
- Recommended capacity (with 20% safety margin): $17.5 \text{ Ah} \times 1.2 =$ **21 Ah**

Extended Operation (4 hours)

- Energy requirement: $210 \text{ W} \times 4 \text{ h} = 840 \text{ Wh}$
- Capacity at 24V: $\frac{840 \text{ Wh}}{24 \text{ V}} = 35 \text{ Ah}$
- Recommended capacity (with 20% safety margin): $35 \text{ Ah} \times 1.2 =$ **42 Ah**

For testing purposes and considering affordability, we will use 9Ah batteries in the current prototype.

6.6 Wheels

6.6.1 Wheel Options

1. **BaneBots Wheel**
2. **Tebru Industrial Robot Wheel**
3. **AGV Drive Wheel**
4. **Modified Heavy Duty Wheel - Robokits India**

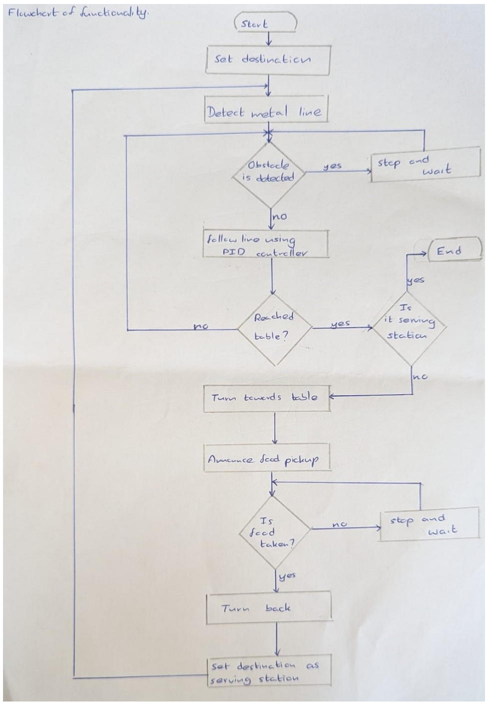
6.6.2 Shaft Hub Options

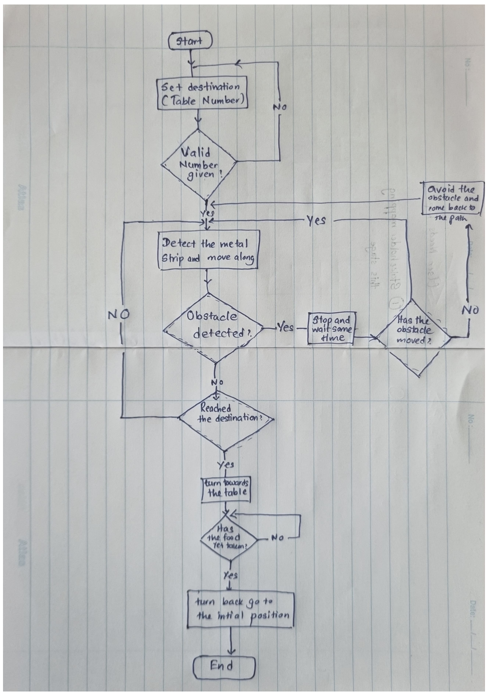
1. **Pololu 8mm Hub**

In conclusion, finding the right wheel for our 8 mm shaft setup has proven challenging due to either compatibility issues or high costs. The best solution appears to be going for a less costly wheel, because of that we chose the Modified Heavy Duty Wheel from Robokits India.

6.7 Flow Chart

6.7.1 Conceptual Design



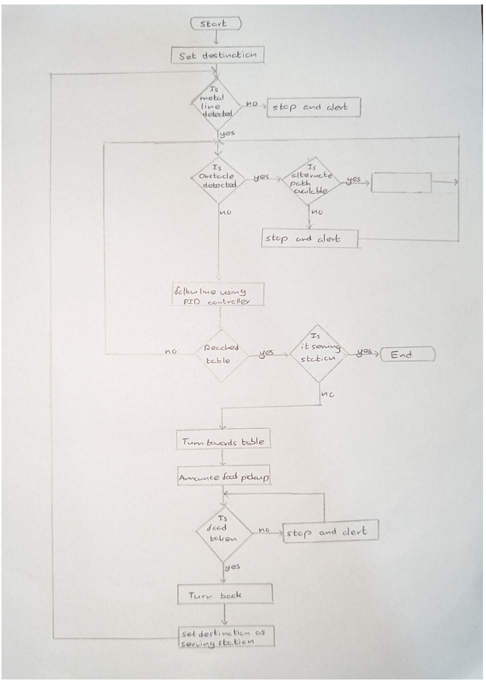


6.7.2 Evaluation

Table 3: Flowchart Evaluation

Criteria	220040T	220061H	220074B	220310X	220452H	220577U	220596C	220663F	220705M	220711D
Logical flow and operation	4	4	4	5	3	3	4	5	3	4
Completeness of process	5	4	3	5	3	4	3	5	3	4
Decision making for obstacles	4	3	3	5	3	3	3	5	3	3
Sensor inputs	4	3	4	5	4	3	4	5	3	3
Customize	4	4	4	5	3	3	3	5	2	3
Overall	21	18	18	25	16	16	17	25	14	17

6.7.3 Final Cross-Pollinated flow chart



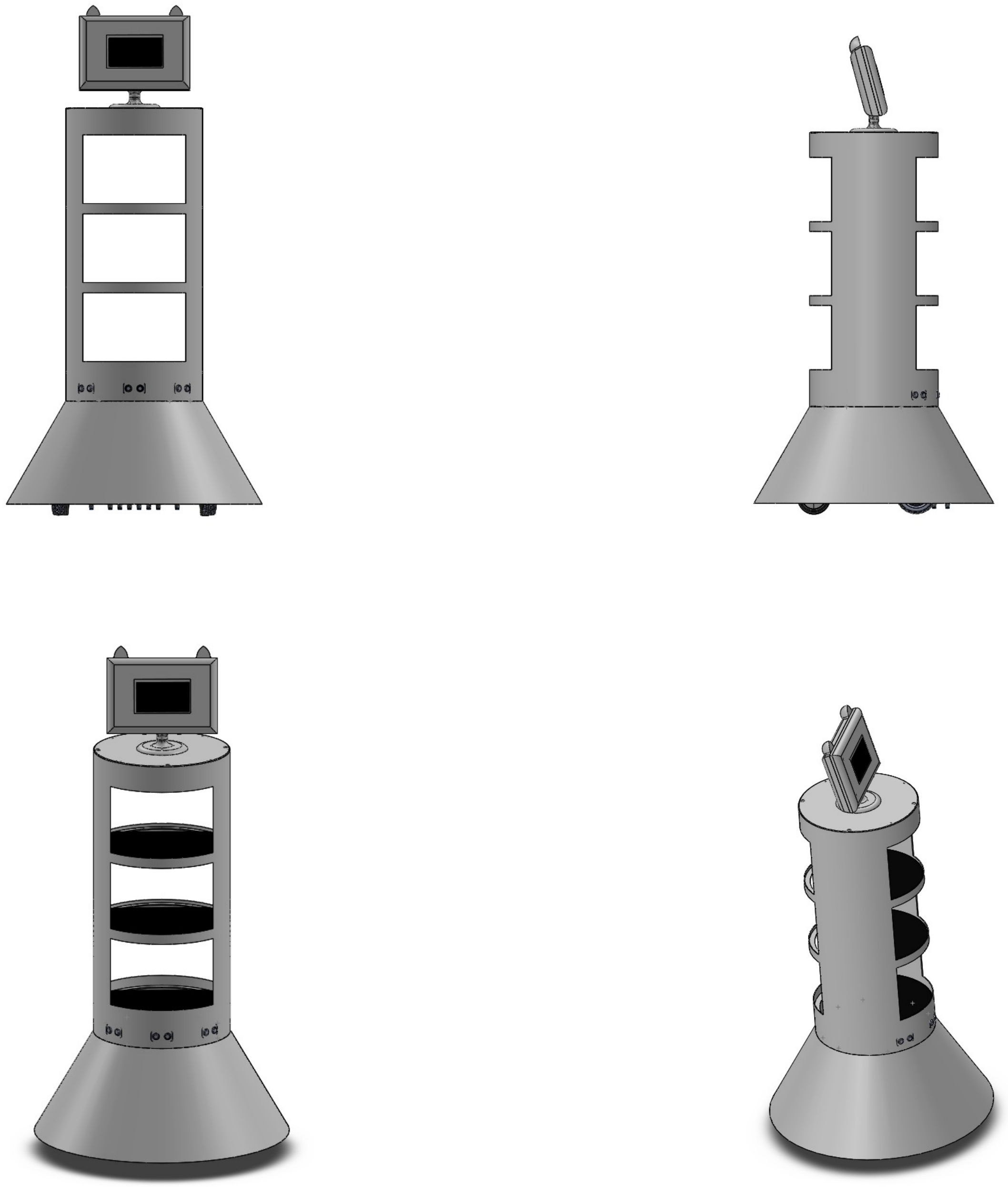
7 New Design Iteration

This section outlines the key improvements and additions made in the latest iteration of the design. Each update was implemented based on evolving project requirements.

8 Updated Mechanical Design

In the latest iteration, the SolidWorks enclosure design was revised to improve manufacturability and cost effectiveness.

8.1 Updated SolidWorks Design



9 Assembled Robot

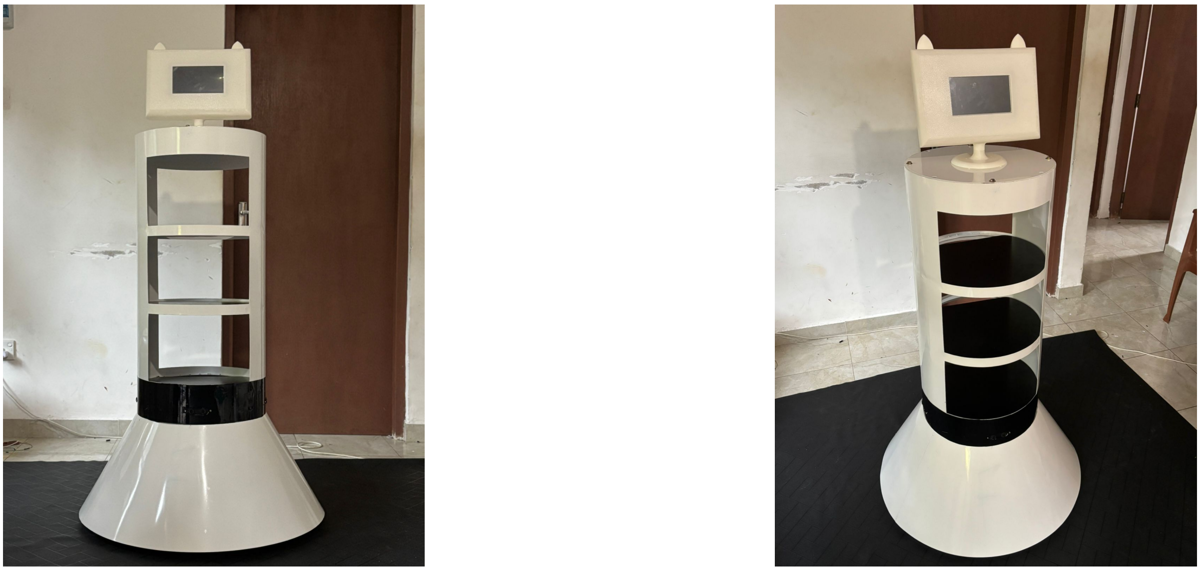


Figure 5: Robot After Assembly

10 Related Calculations

10.1 Mathematical Equation for Acceleration S-Curve

The acceleration velocity profile of the robot is modeled using a sigmoid function. This approach enables smooth transitions from rest to the maximum velocity $V_{\max}$, avoiding abrupt acceleration spikes that could affect mechanical stability or user comfort. The velocity function $v(t)$ over time is defined as:

$$v(t) = \frac{V_{\max}}{1 + e^{-K(t - X_0)}}$$

Where:

- $v(t)$ — velocity at time t
- $V_{\max}$ — maximum velocity (e.g., 50.0)
- K — steepness factor that controls how fast acceleration ramps up (positive constant)
- X_0 — midpoint of the sigmoid curve (i.e., the time when velocity reaches $\frac{V_{\max}}{2}$)

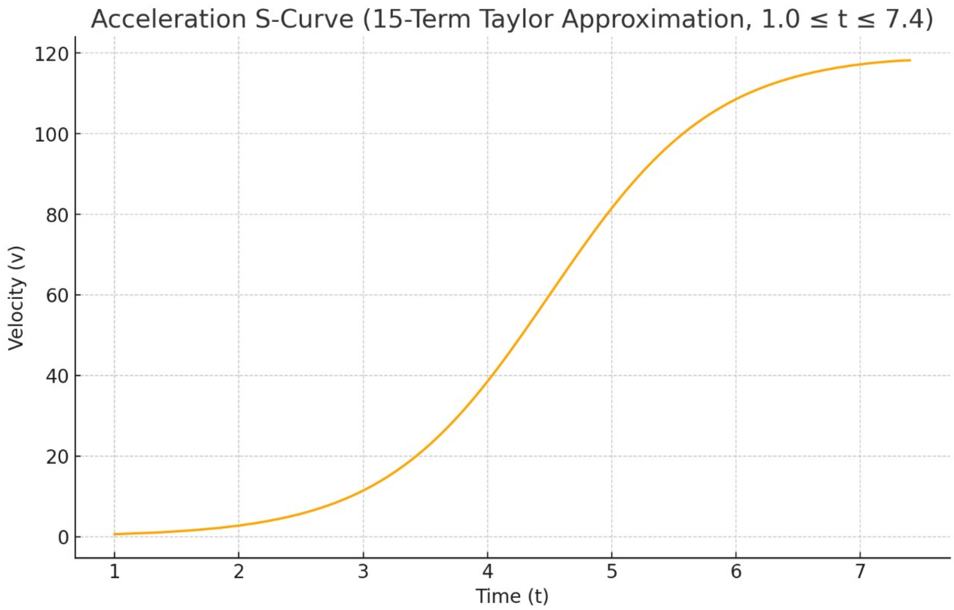


Figure 6: Acceleration S-Curve

10.2 Mathematical Equation for Deceleration S-Curve

For deceleration, a reverse sigmoid function is used to gradually reduce the velocity from $V_{\max}$ to near zero. This ensures the robot stops smoothly, minimizing jerk and maintaining control.

$$v(t) = \frac{V_{\max}}{1 + e^{K(t-X_0)}}$$

The parameters K and X_0 retain the same definitions as in the acceleration equation, with $K > 0$ to reverse the sigmoid curve shape.

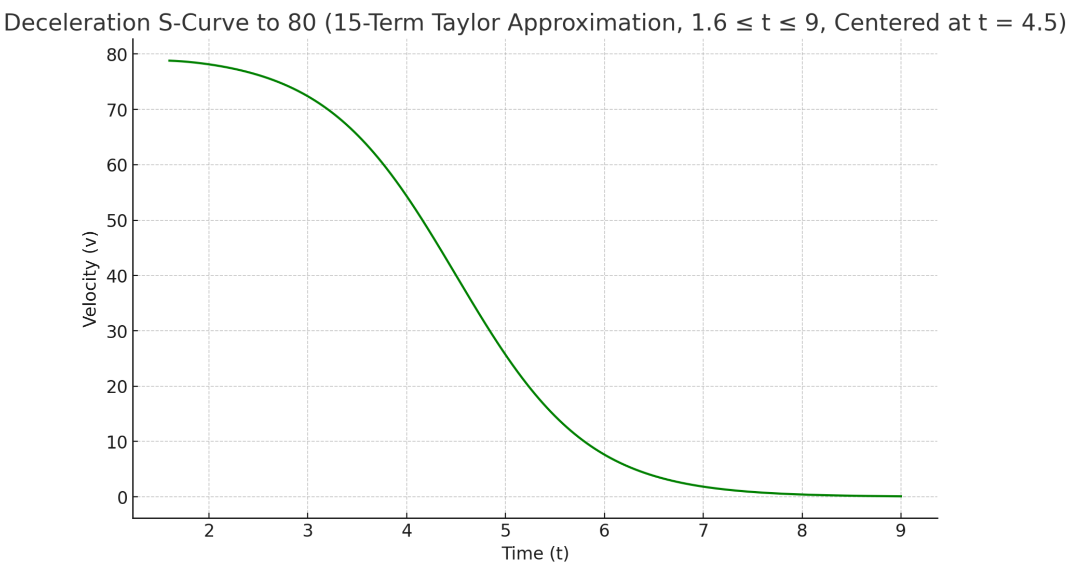


Figure 7: Decelaration S-Curve

10.3 Use in Motion Planning

These equations were used to program smooth acceleration and deceleration phases in the robot's motor control logic. By adjusting the parameters K and X_0, the team could fine-tune how quickly or gradually the robot transitions between different speeds, enabling efficient, safe, and comfortable movement across the restaurant floor.

11 Power Supply Design

The power supply system was designed to provide stable and efficient voltage levels required by various subsystems of the project. A rechargable LiPo battery pack was selected as the primary power source due to its high energy density, lightweight nature, and ability to deliver high discharge currents suitable for motor drivers and processing units.

11.1 Power Regulation Architecture

The output from the LiPo battery is first boosted to 24V using a high-efficiency DC-DC boost converter. This 24V rail serves as the main power bus for high-power components such as motor drivers. Subsequently, the 24V line is stepped down to 12V using a buck converter to power intermediate subsystems.

Within the main PCB, the 12V supply is further regulated to 5V and 3.3V using linear and switching regulators to supply logic-level components, sensors, and microcontrollers. This tiered voltage regulation strategy ensures optimal power distribution, minimizes power losses, and isolates noise-sensitive components from high-current lines.

12 PCB Design

12.1 Main PCB Design

A 4-layer PCB was designed to meet the signal integrity, power distribution, and compact layout requirements of the system. This multi-layer structure allows for better separation of power and signal planes, reduced electromagnetic interference, and efficient routing.

12.2 Layer Stackup

The printed circuit board (PCB) for the robot was designed as a 4-layer stack to ensure signal integrity, reduce electromagnetic interference (EMI), and support high-current power delivery. The layer configuration is as follows:

Table 8: PCB Layer Stackup

Layer	Description
Top Layer	Signal routing
Layer 2	Ground plane
Layer 3	Power plane and additional ground zones
Bottom Layer	Signal routing

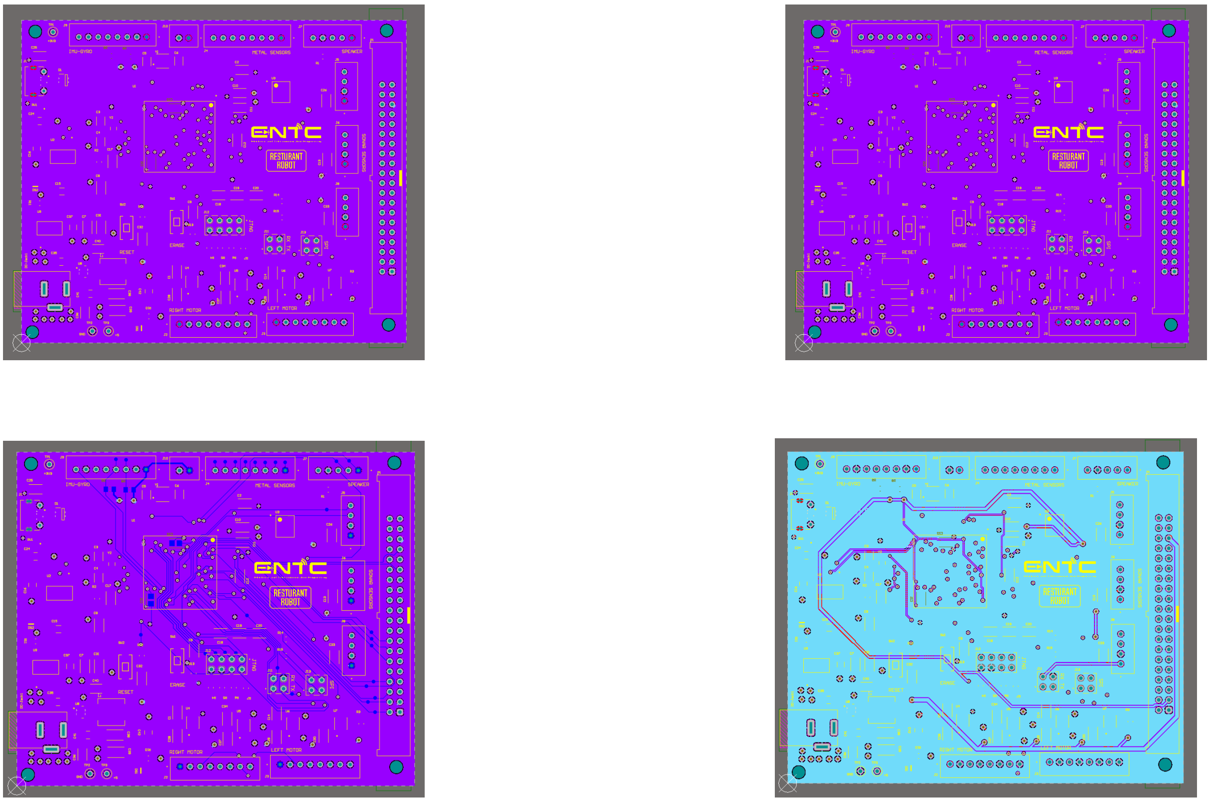


Figure 8: Layer-wise PCB Views

12.3 Key Considerations

The following key considerations were taken into account during the PCB design and fabrication process:

- **Wide Power Traces and Polygon Pours:** Power traces were made wider and polygon pours were applied to high-current paths to minimize voltage drop and thermal resistance, thereby improving overall power integrity and heat dissipation.
- **Strategic Component Placement:** Components were placed to ensure optimal signal flow, minimize trace lengths, and reduce noise. Special attention was given to the placement of sensitive analog and high-frequency components.

- **Voltage Regulator Placement:** The voltage regulator was positioned in accordance with the layout guidelines provided in its datasheet, ensuring proper thermal management and reliable voltage regulation.
- **Differential Pair Routing:** Differential signal lines were routed using matched-length traces with controlled impedance. This reduces electromagnetic interference (EMI) and ensures signal integrity for high-speed communication.
- **Test Point Provisioning:** Dedicated test points were added at critical locations across the board to facilitate debugging, signal probing, and quality control testing. This enhances the maintainability and testability of the final product.

12.4 PCB

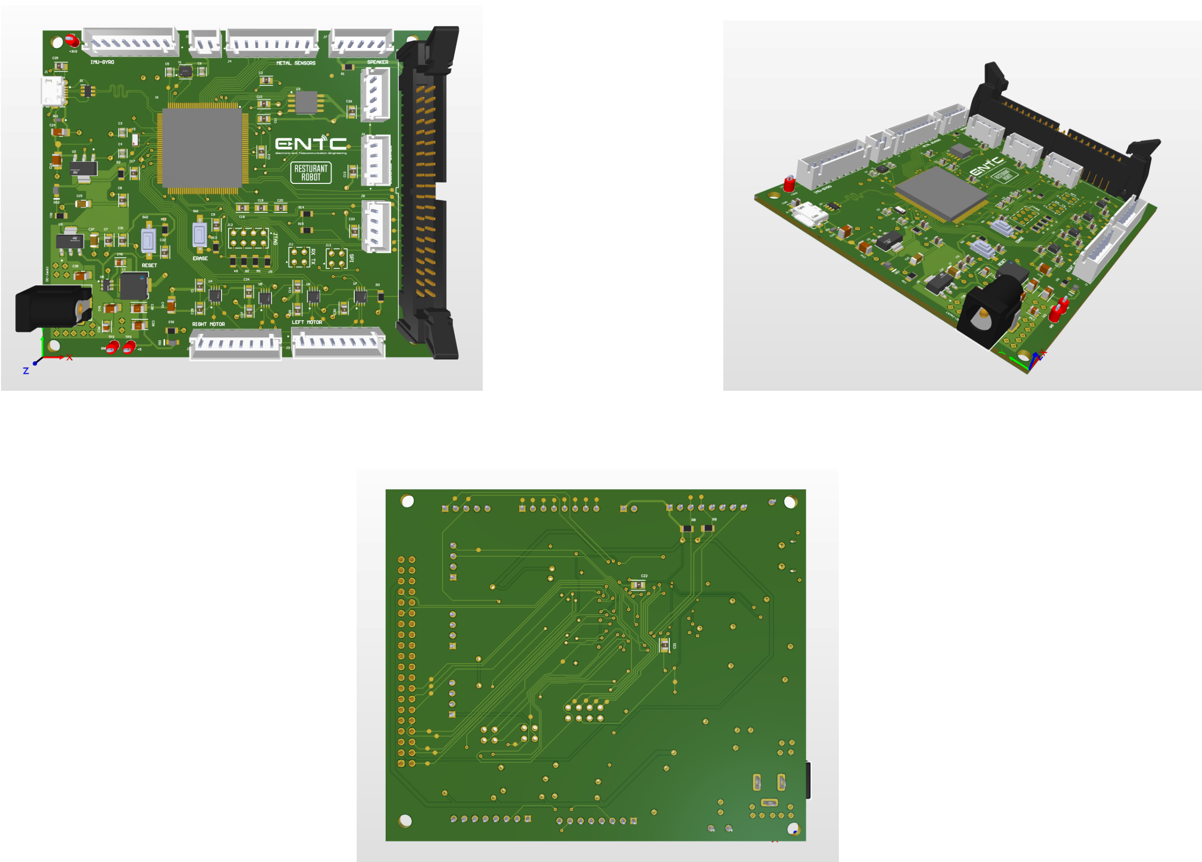
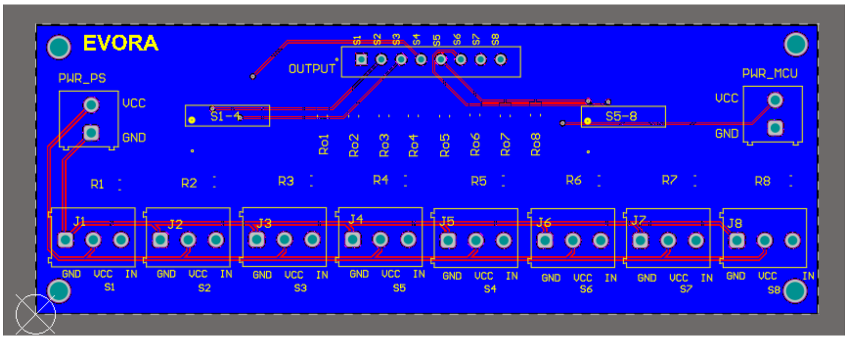


Figure 9: Final PCB Renderings

12.5 Sensor Array PCB Design

In addition to the main control board, a separate PCB was designed specifically for the ultrasonic sensors. This modular approach allows for easier maintenance, signal isolation, and optimized sensor placement within the mechanical enclosure.



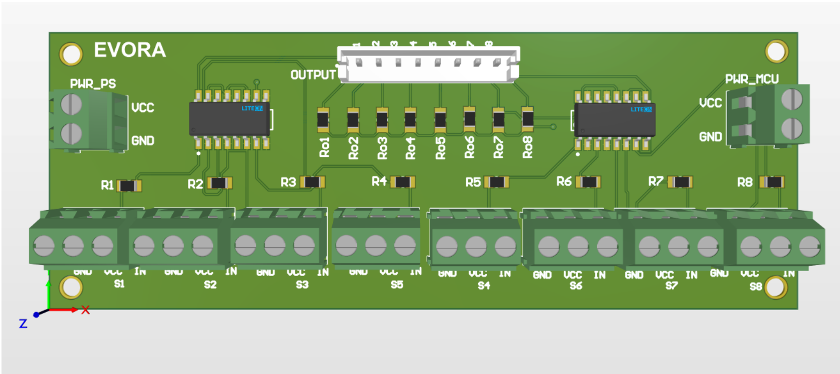


Figure 10: PCB for the Proximity Sensor array

13 Firmware Development

The firmware for the restaurant robot was developed in C using the Atmel Software Framework (ASF) and deployed on the SAM3X8E microcontroller. It manages key functionalities such as line following, obstacle detection, motor control, display updates, and inter-module communication.

13.1 Modular Architecture

The firmware is divided into several functional modules, each responsible for a critical subsystem. These include:

- **Motor Control:** PWM signals are generated to drive two NEMA 23 closed-loop stepper motors. Acceleration and deceleration are managed using S-curve velocity profiles for smooth motion.
- **Sensor Handling:** Data from proximity and ultrasonic sensors are continuously read and processed to detect obstacles and junctions for decision-making during navigation.

- **Line Following Algorithm:** A PID control algorithm calculates error based on sensor input and adjusts the motor speeds to follow metal paths embedded under the floor.
- **S-Curve Velocity Profiling:** Acceleration and deceleration curves are implemented using sigmoid functions to ensure smooth speed transitions and mechanical stability.
- **Display Interface:** The 7" TFT LCD is controlled via a parallel interface to display order status and user interaction elements. Display logic is separated from core navigation code for modularity.
- **Obstacle Response:** In case of detected obstacles, the robot executes a controlled deceleration sequence and makes path decisions based on junction detection logic.

14 RS-232 Communication Integration

RS232 communication was integrated into the system to enable reliable serial communication between microcontrollers (MCUs), particularly for cases where asynchronous, long-distance, or noise-tolerant data transmission is required. The MAX3232 transceiver module was used to convert TTL-level UART signals from the microcontroller to RS232-compliant voltage levels, allowing seamless interfacing with devices that operate on RS232 standards.

This module was chosen for its dual-channel support, low power consumption, and compatibility with a wide operating voltage range, making it ideal for embedded applications. A suitable DB9-compatible connector was included on the PCB for secure and standardized physical interfacing.

By incorporating RS232 communication, the design supports inter-MCU coordination, modular subsystem development, and future expandability with external debugging tools or legacy RS232-based controllers.

15 Final Design Iteration - Overview

This section presents the updates and changes implemented in the final design iteration of the restaurant robot. Throughout this stage, the team focused on refining critical subsystems to enhance the robot's functionality and integration for deployment-ready performance. This final design iteration includes the replacement of the display module, redesign of the robot head, finalization of the power supply, and integration of custom PCBs into the robot to replace earlier development boards.

16 Block Diagram

This section presents the final block diagram of the restaurant robot, reflecting all system updates and integrations made in the final design iteration.

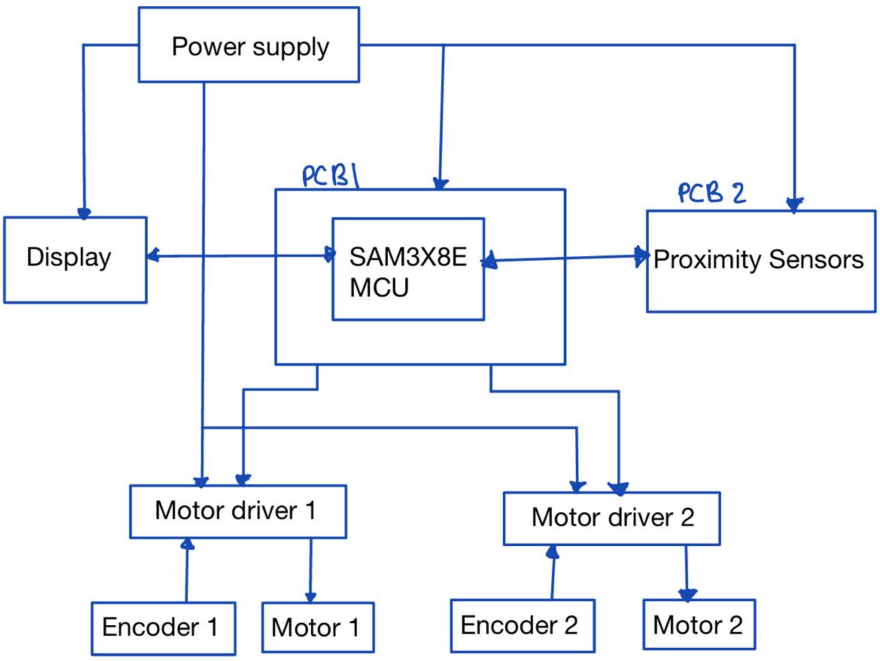


Figure 11: Final Block Diagram

17 Power Supply

In the final iteration, an updated power supply architecture was implemented to ensure stable operation of all subsystems. An 11.1V LiPo battery is used as the main power source. This voltage is boosted to 24V using a dedicated boost converter module, which supplies power to the two motor drivers and the ultrasonic sensor array. Additionally, the 11.1V battery voltage is bucked down to 9V using a buck converter module, and this 9V supply is provided to the main PCB. Within the PCB, onboard regulators are used to further regulate the voltage to 3.3V and 5V as required by the microcontroller, display module, and other electronic components.

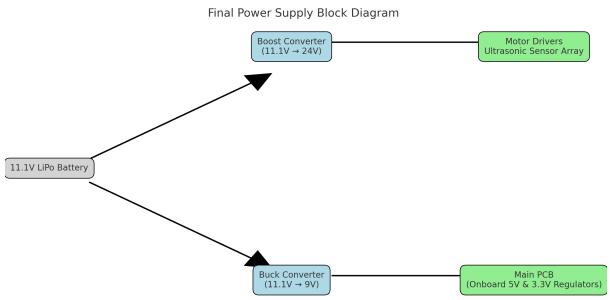


Figure 12: Final Power Supply Block Diagram

18 Display

In the final design iteration, the initial 7-inch TFT LCD display was damaged due to an unexpected electrical issue that caused it to burn out. As a result, the team had to move on to a smaller display module due to budget constraints. The replacement display used was a TFT LCD Shield 2.4" Socket Touch Panel Module. Although smaller in size, this display provided adequate functionality for the robot's interface requirements.

The touch display allows users to choose the table to be served and to turn on or off the googly eyes of the robot. Initially, the googly eyes were planned to be integrated graphically within the 7-inch display interface. However, due to the change in display size and resolution, the team implemented the googly eyes using two 8×8 LED matrices, which effectively emulate two animated eyes on the robot's head.

Figure 13: Touch Panel Module

Figure 14: LED Matrix

19 Enclosure Design

In the final design iteration, the enclosure design was updated only for the robot head to accommodate the change in display module. The head was redesigned to fit the smaller 2.4” TFT LCD touch display along with the two 8×8 LED matrices used for the googly eyes, ensuring proper integration, aesthetics, and structural stability.

19.1 Updated Sketch

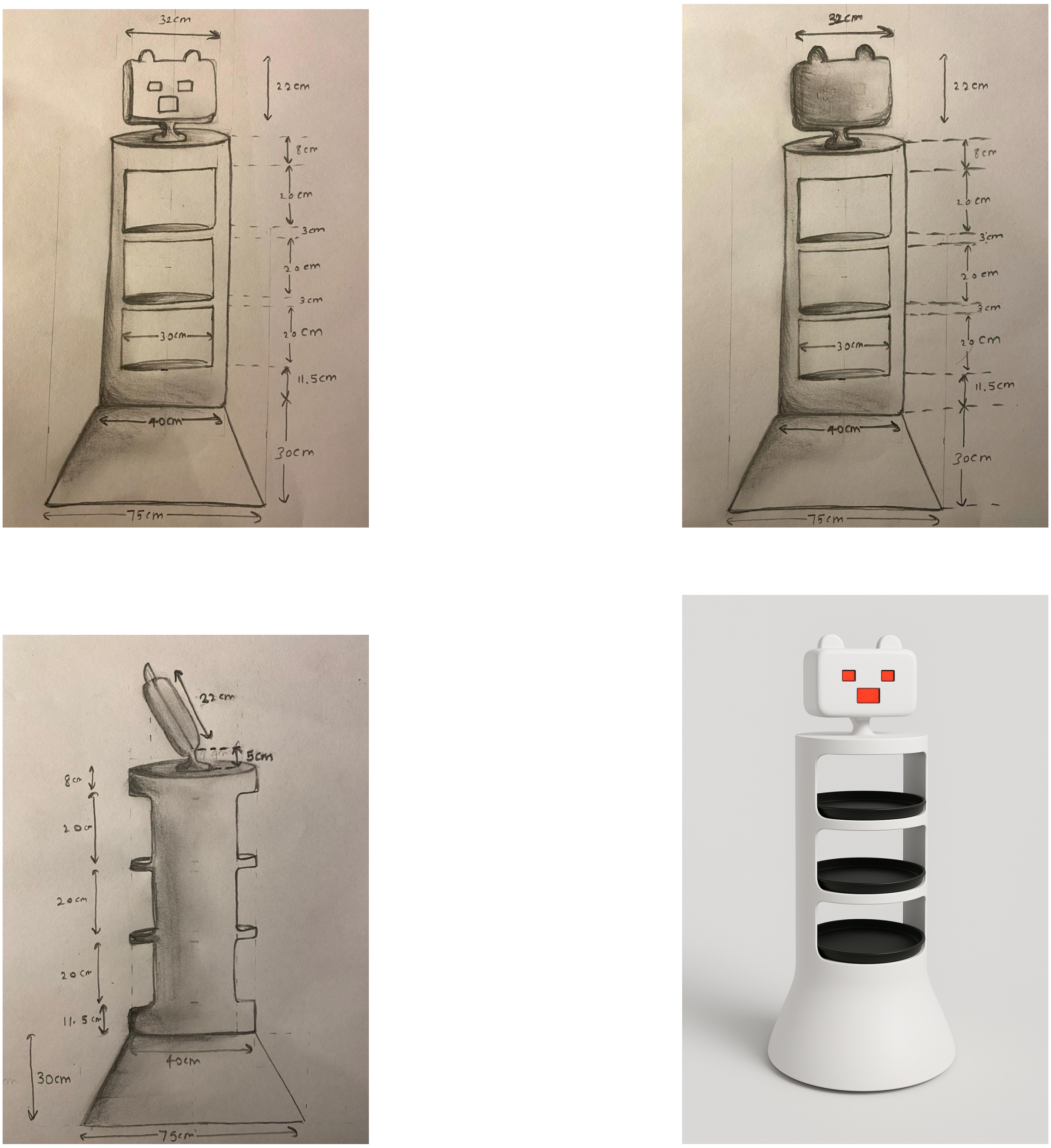


Figure 15: Updated sketches

19.2 Updated Solidworks Design

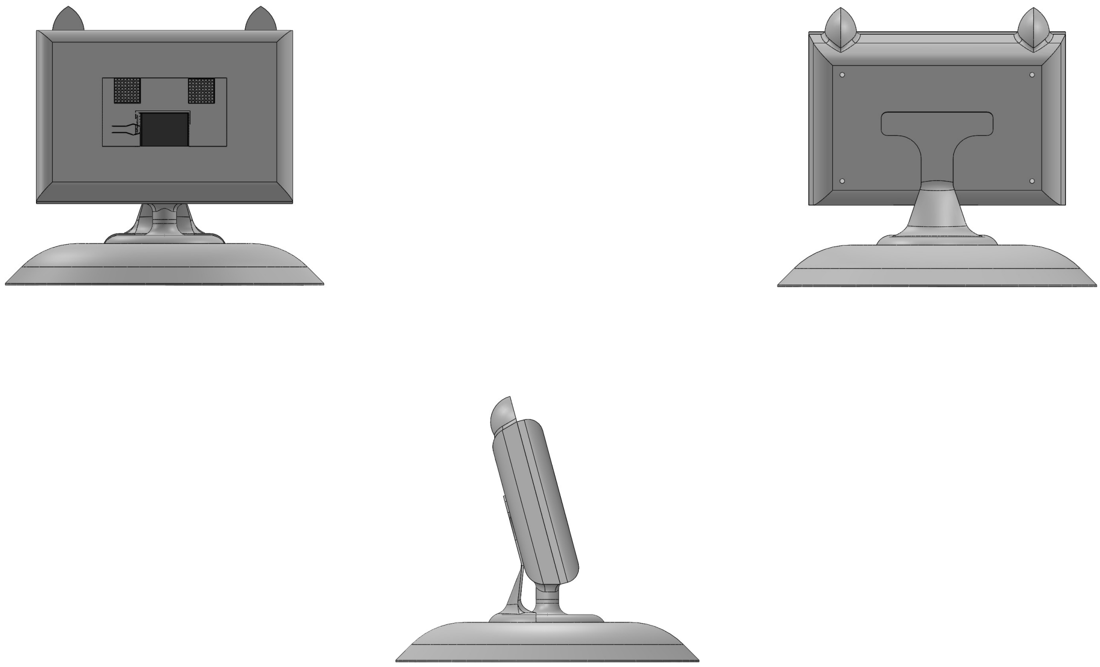


Figure 16: Updated robot head design

19.3 Assembled Robot

Figure 17: Assembled Robot

20 PCB

In the final design iteration, custom PCBs were designed and integrated into the robot, replacing the earlier use of development boards. During the last evaluation, the PCB had not yet been shipped to the team. After receiving the PCB, the team assembled and thoroughly debugged it, ensuring functionality and integration with all subsystems. This upgrade improved electrical robustness, reduced wiring complexity, and optimized space utilization within the mechanical structure.

21 PCB Pin Allocation

Table 9 shows the pin allocation of the main PCB in the final design. The allocation includes motor control pins, metal detecting sensor inputs, ultrasonic sensor connections, and power lines.

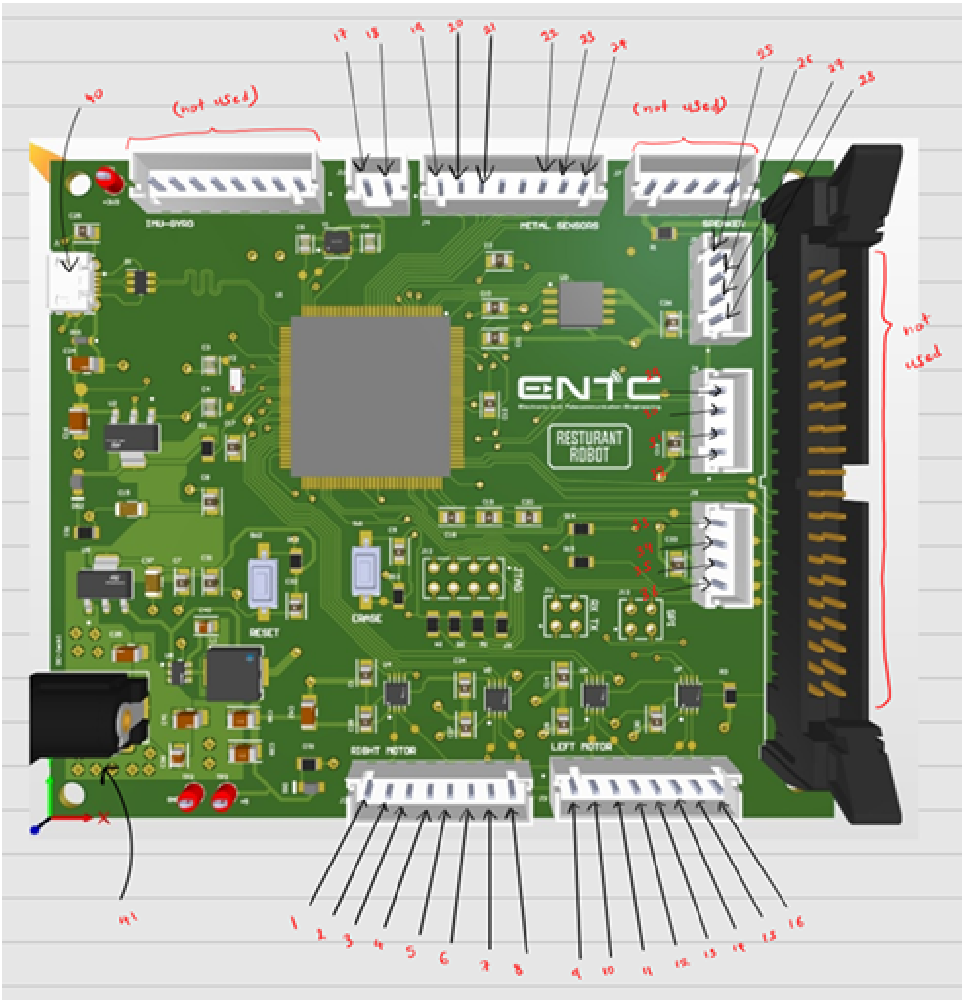


Figure 18: Pin Allocation

Pin	Function
1	RM PU
2	GND
3	RM DR
4	GND
5	RM MF
6	GND
7	RM ALM
8	GND
9	LM PU
10	GND
11	LM DR
12	GND
13	LM MF
14	GND
15	LM ALM
16	GND
17	GND
18	3.3V
19-24	Metal detecting 1-8
25	PWM SONAR 1
26	TRG SONAR 1
27	GND
28	3.3V
29	PWM SONAR 2
30	TRG SONAR 2
31	GND
32	3.3V
33	PWM SONAR 3
34	TRG SONAR 3
35	GND
36	3.3V

Table 9: PCB pin allocation summary for the restaurant robot.

21.1 Bare and Assembled PCB

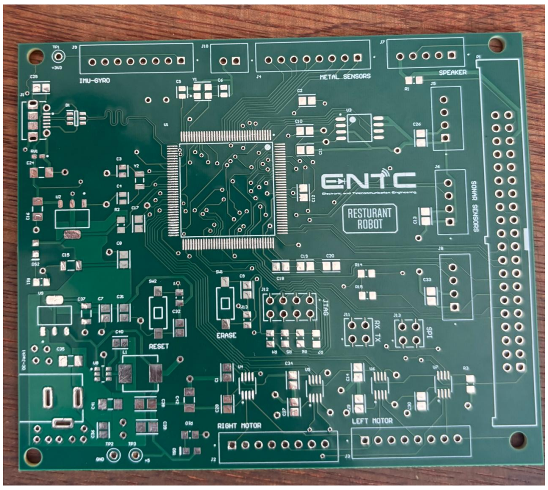


Figure 19: Bare Main PCB

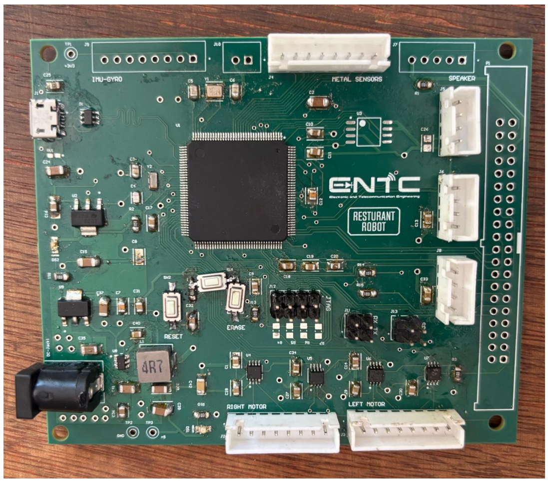


Figure 20: Assembled Main PCB

21.2 Important Considerations and Changes due to Practical issues

Several important considerations were addressed in the final PCB implementation:

- **Ultrasonic Sensor Array Spacing:** In the ultrasonic sensor array PCB, only 6 out of the 8 headers were used. This was done to reduce the number of sensors to six, allowing sufficient spacing between adjacent sensors to minimize cross-interference and ensure reliable obstacle detection.

- **IMU and Speaker System Exclusion:** The IMU and speaker system were not implemented in the final design due to time constraints and the complexity involved in their integration within the available project timeline.

- **SWD (Serial Wire Debug) Interface:** The SWD interface is implemented for programming and real-time debugging of the microcontroller. We are Using the J-Link debugger, to upload firmware efficiently and perform in-circuit debugging.

- **RS-232 Interface:** RS-232 communication is used for serial data exchange between the display and our main controller PCB. This interface is used for debugging, logging, and could be extended to connect to other RS-232-compatible modules. The RS-232 signals are level-shifted appropriately to ensure compatibility with external devices.

22 Acceleration Curve

There were no changes made to the acceleration curve design in the final iteration. The same S-curve profile was used to ensure smooth acceleration and deceleration of the robot.

```
//S-curve
// VMAX=75.0 , K=1.5 , X0=4.5
float calculate_s_curve_velocity_acceleration(float t) {
  float exponent = -K * (t - X0); // Notice the minus here
  float exp_value = expf(exponent);
  float v = VMAX / (1.0f + exp_value);

  return v;
} for (float t = 3.0f; t <= 8.0f; t += 0.1f) {
     Velocity = calculate_s_curve_velocity_acceleration(t);
     frequency = 40 * Velocity;
     processLineFollowing();
     //mdrive((40*Velocity),(40*Velocity));
            delay_ms(25); }
```

Listing 1: Acceleration Curve Code

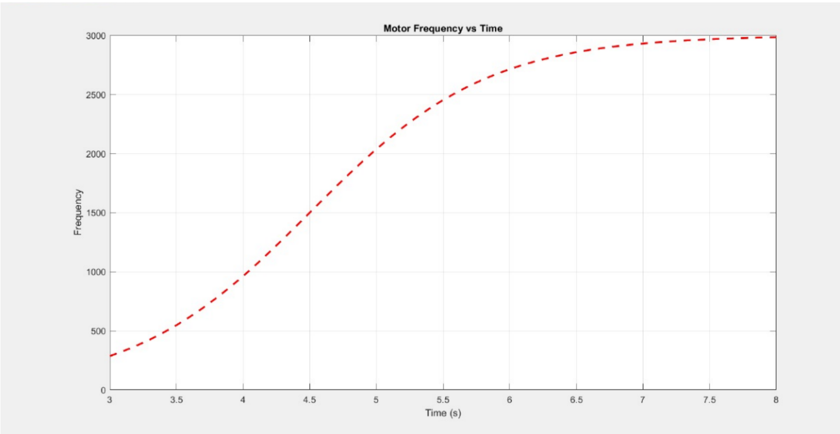


Figure 21: Acceleration curve

```
//S-curve
// VMAX=75.0 , K=1.5 , X0=4.5
float calculate_s_curve_velocity_acceleration(float t) {
  float exponent = -K * (t - X0); // Notice the minus here
  float exp_value = expf(exponent);
  float v = VMAX / (1.0f + exp_value);

  return v;
} for (float t = 3.0f; t <= 8.0f; t += 0.1f) {
     Velocity = calculate_s_curve_velocity_acceleration(t);
```

```
    frequency = 40 * Velocity;
    processLineFollowing();
    //mdrive((40*Velocity),(40*Velocity));
          delay_ms(25); }
```

Listing 2: Deceleration Curve Code

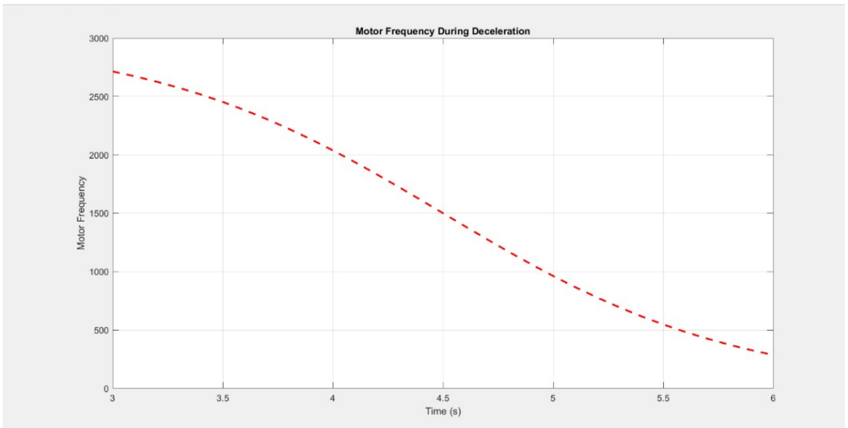


Figure 22: Deceleration curve

23 Firmware

Changes were made to the firmware in the final iteration to integrate the updated hardware components and display module. The updated code is provided in the project code base for reference.

24 Conclusion

The final iteration of the restaurant robot design represents a significant advancement in system integration, mechanical refinement, and electronic reliability. By replacing development boards with custom-designed PCBs, the team enhanced the robot's stability, compactness, and wiring simplicity. Critical updates—including a power supply architecture, an efficient display solution using a 2.4" TFT touchscreen and 8×8 LED matrices, and improved mechanical enclosure design—enabled the robot to function reliably in its intended environment.

Despite encountering practical limitations such as display damage, sensor spacing issues, and the exclusion of the IMU and speaker system, the team successfully adapted and optimized the robot within available time and resource constraints. The continued use of the S-curve acceleration profile ensured smooth and controlled motion, essential for a service robot operating in restaurant spaces.

Overall, the project demonstrates a strong application of electronic design realization principles and effective problem-solving under real-world constraints. The outcome is a functional, modular, and well-documented restaurant robot prototype that lays the groundwork for future improvements and potential deployment in hospitality environments.

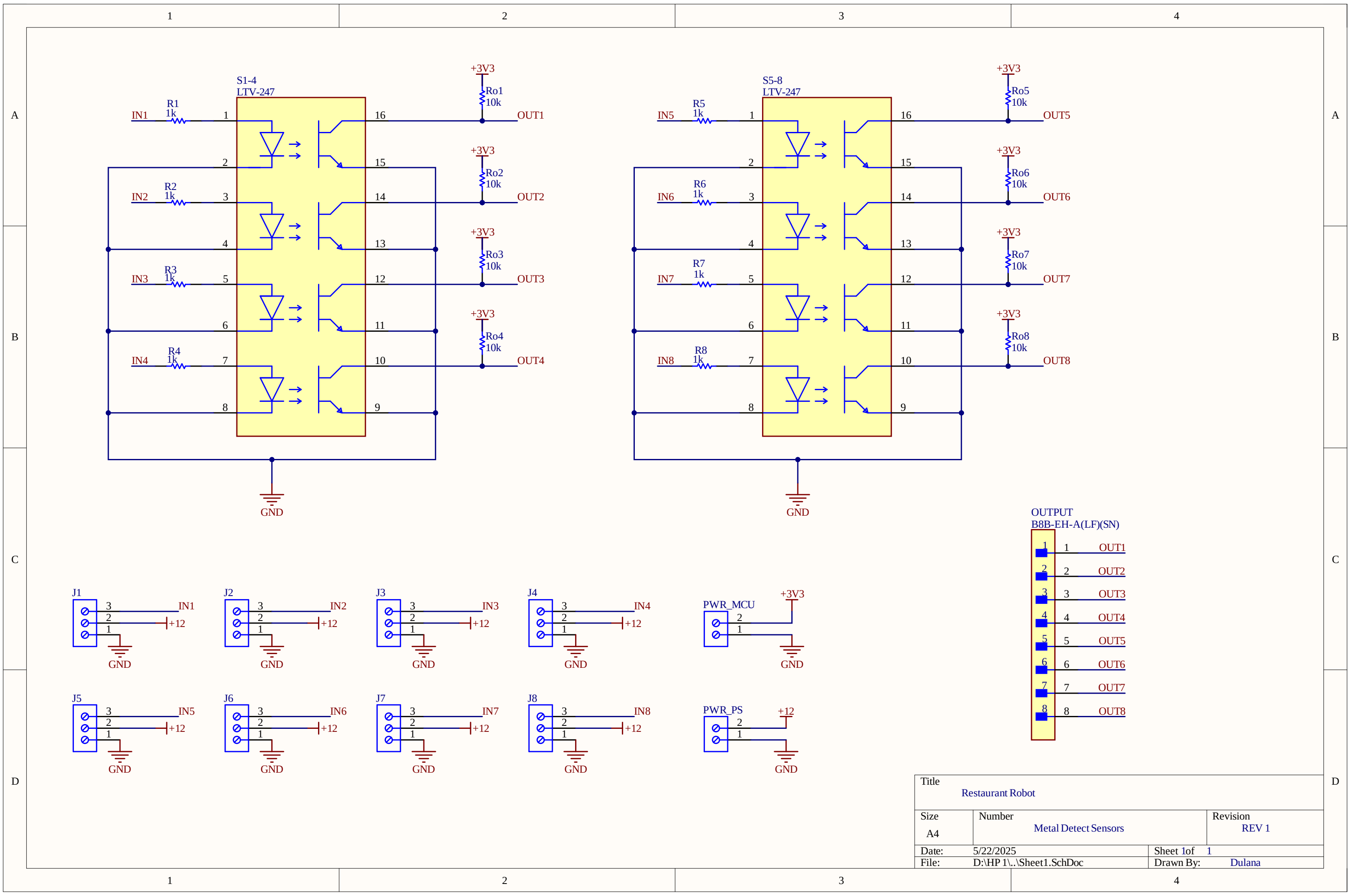
A Appendices

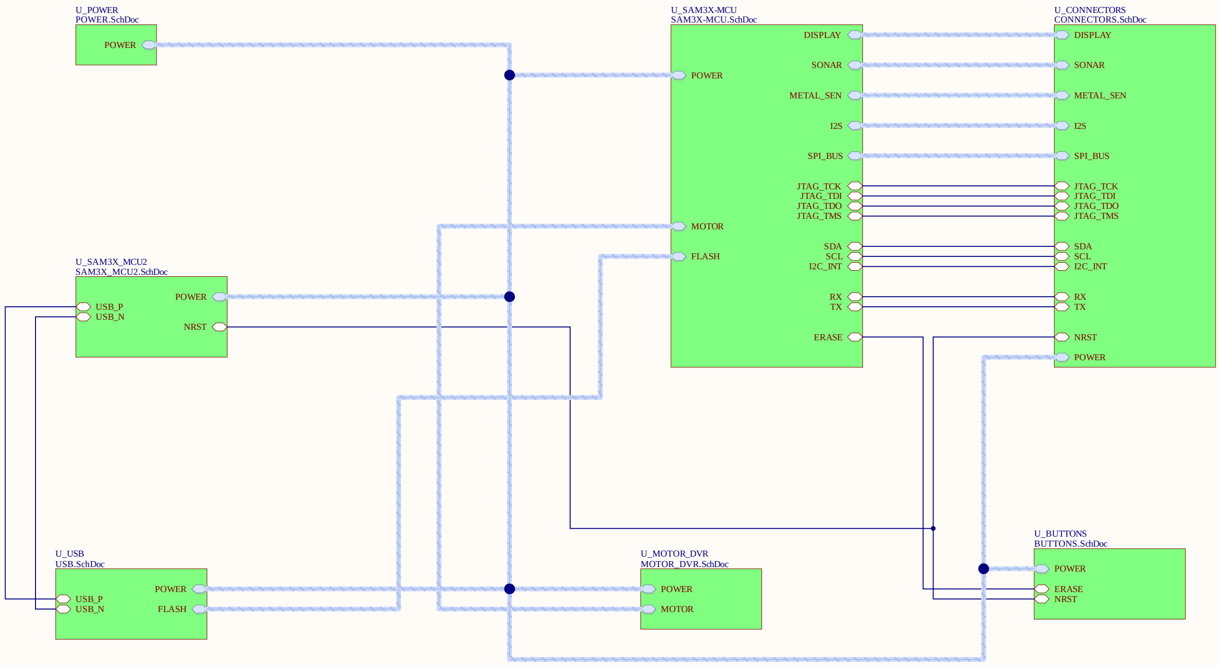
A.1 Documents

- **Nema 23 Motor and Driver**:https://drive.google.com/drive/folders/1t_PNO2JcCgiCsATQyiluq-kjOZTv335-?usp=drive_link
- **AO2 PWM Ultrasonic Sensors**: https://drive.google.com/drive/folders/1_Xdv5ZsOF8AB279o9QYAdrCsSYbcHOdw?usp=drive_link
- **Proximity Sensors**: https://drive.google.com/drive/folders/1keqV7BNKRRuUt0pEe417E9VMuuxMAJQu?usp=drive_link
- **7-inch TFT LCD Display**: https://drive.google.com/drive/folders/14D8buxf1hrKpHv2f4uz1sQXVDi9iKZ4L?usp=drive_link
- **SAM3X8E MCU**: https://drive.google.com/file/d/1gG4gJJS6T9XMbzaYF70B0jIM091cdNFk/view?usp=drive_link
- **Wheel**: https://drive.google.com/drive/folders/1SpT7xqD51tSb7tsyiDhPFohJYaKiCV0P?usp=drive_link

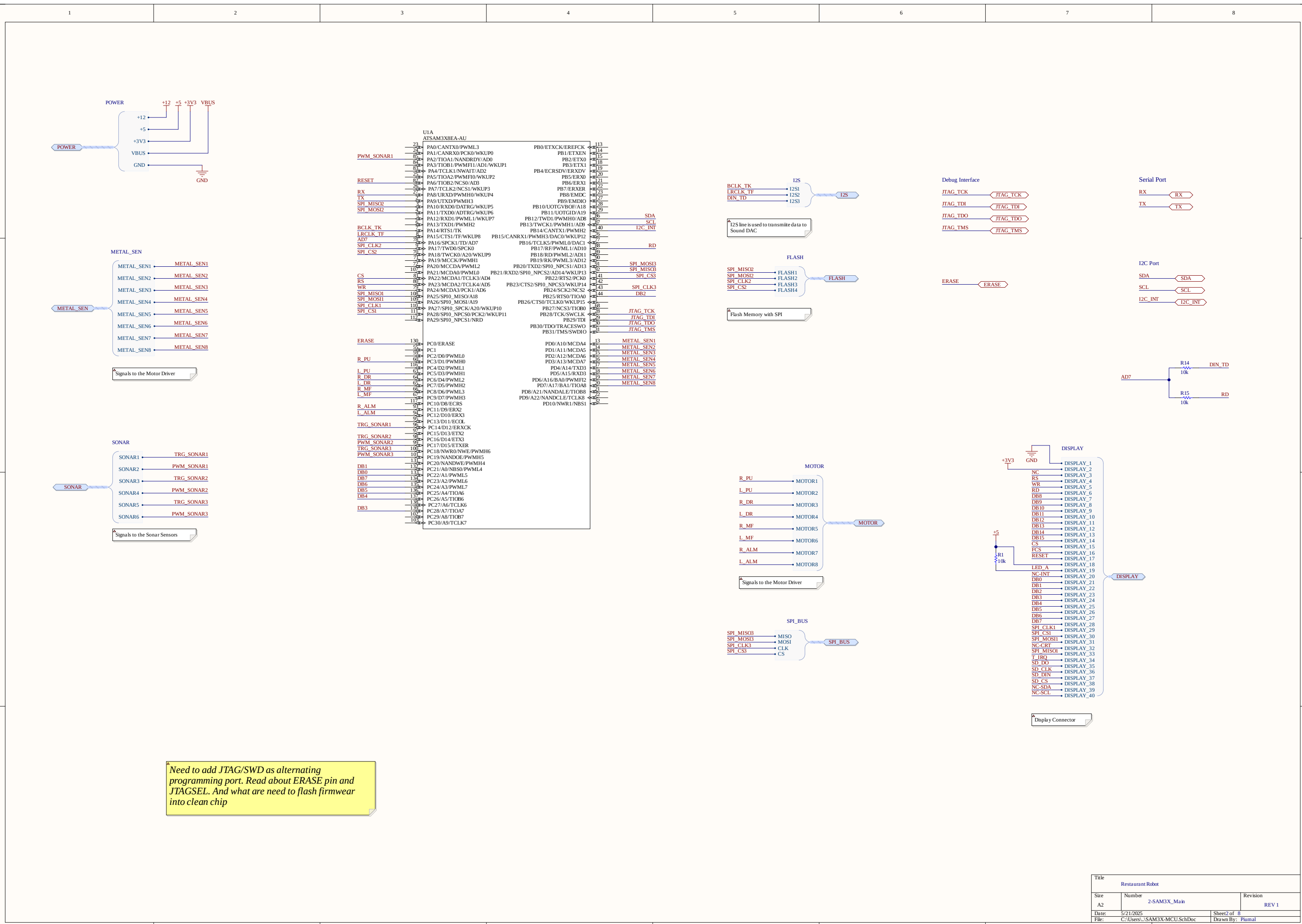
A.2 Full Schematic Diagrams

This section contains the schematic diagrams of the 2 PCBs, illustrating the interconnections between all major components and subsystems used in the restaurant robot.


S1-4
LTV-247
S5-8
LTV-247
IN1 R1 1k
IN2 R2 1k
IN3 R3 1k
IN4 R4 1k
IN5 R5 1k
IN6 R6 1k
IN7 R7 1k
IN8 R8 1k
+3V3 Ro1 10k OUT1
+3V3 Ro2 10k OUT2
+3V3 Ro3 10k OUT3
+3V3 Ro4 10k OUT4
+3V3 Ro5 10k OUT5
+3V3 Ro6 10k OUT6
+3V3 Ro7 10k OUT7
+3V3 Ro8 10k OUT8
GND
OUTPUT
B8B-EH-A(LF)(SN)
J1 IN1 +12 GND
J2 IN2 +12 GND
J3 IN3 +12 GND
J4 IN4 +12 GND
J5 IN5 +12 GND
J6 IN6 +12 GND
J7 IN7 +12 GND
J8 IN8 +12 GND
PWR_MCU +3V3 GND
PWR_PS +12 GND
Title
Restaurant Robot
Size A4
Number Metal Detect Sensors
Revision REV 1
Date: 5/22/2025
Sheet 1of 1
File: D:\HP 1\..\Sheet1.SchDoc
Drawn By: Dulana

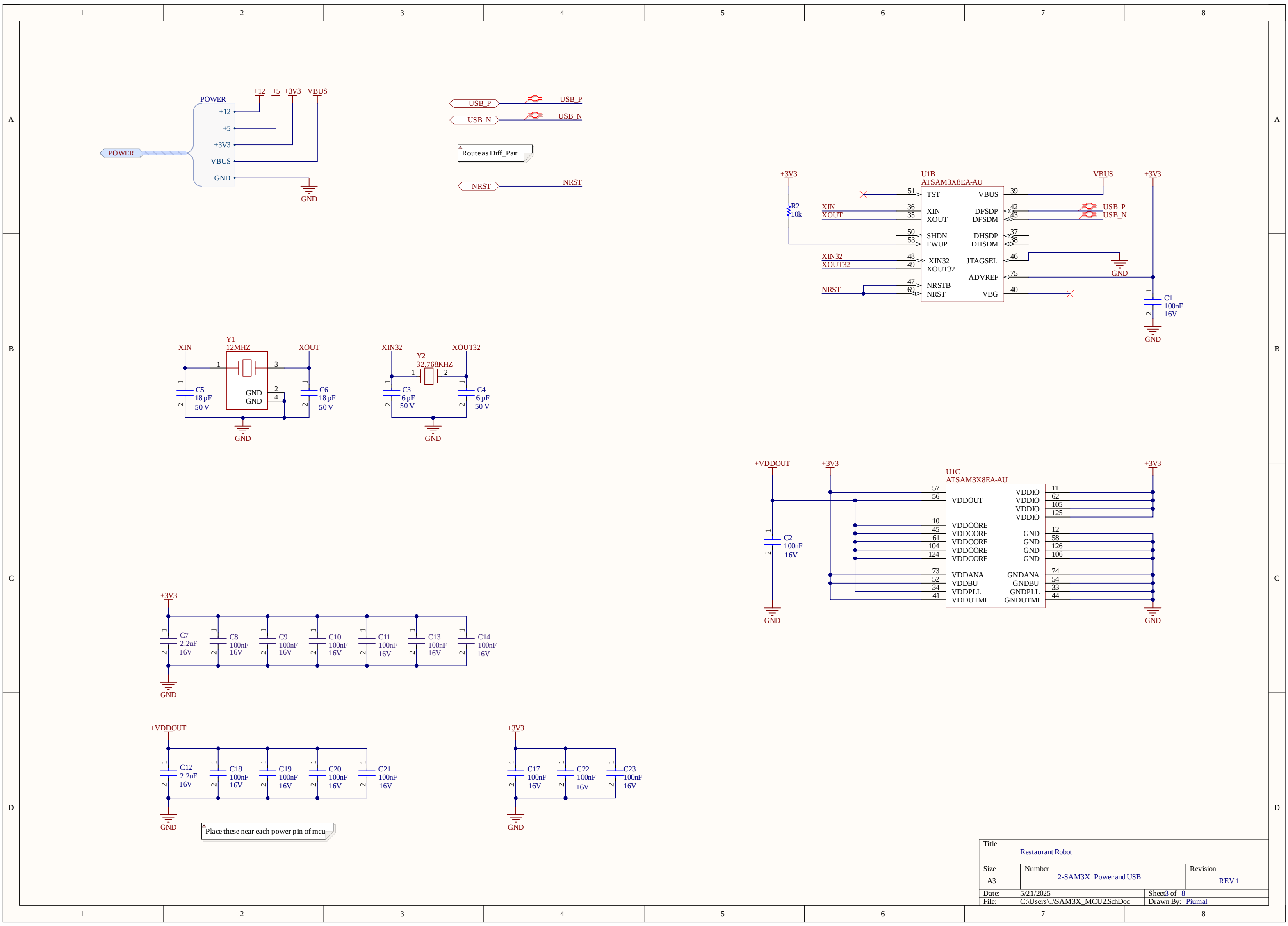


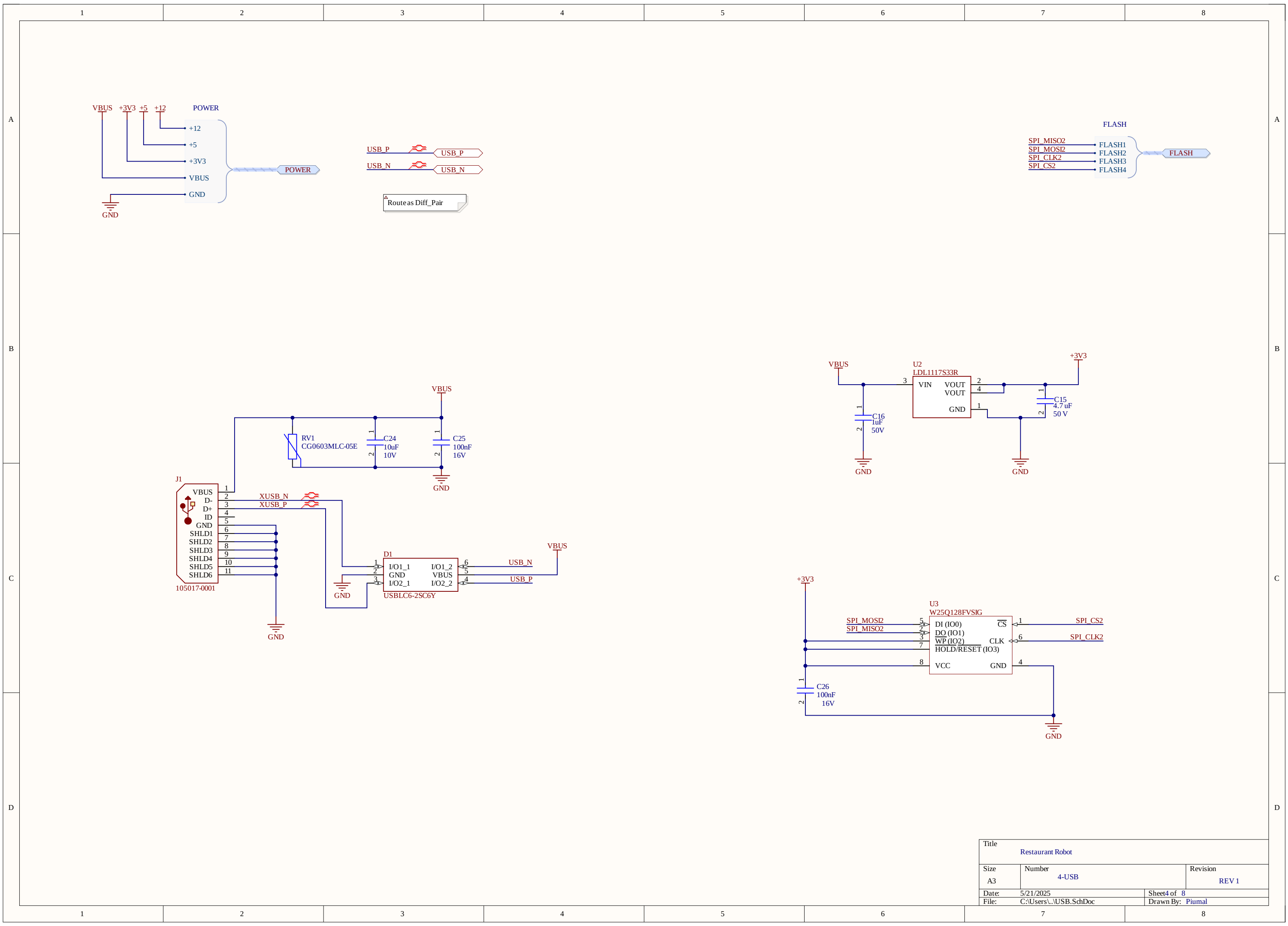
Title	Restaurant Robot		
Size A3	Number 1-Top		Revision REV 1
Date: 5/21/2025		Sheet 1 of 8	
File: C:\Users\..\Top.SchDoc		Drawn By: Piumal	



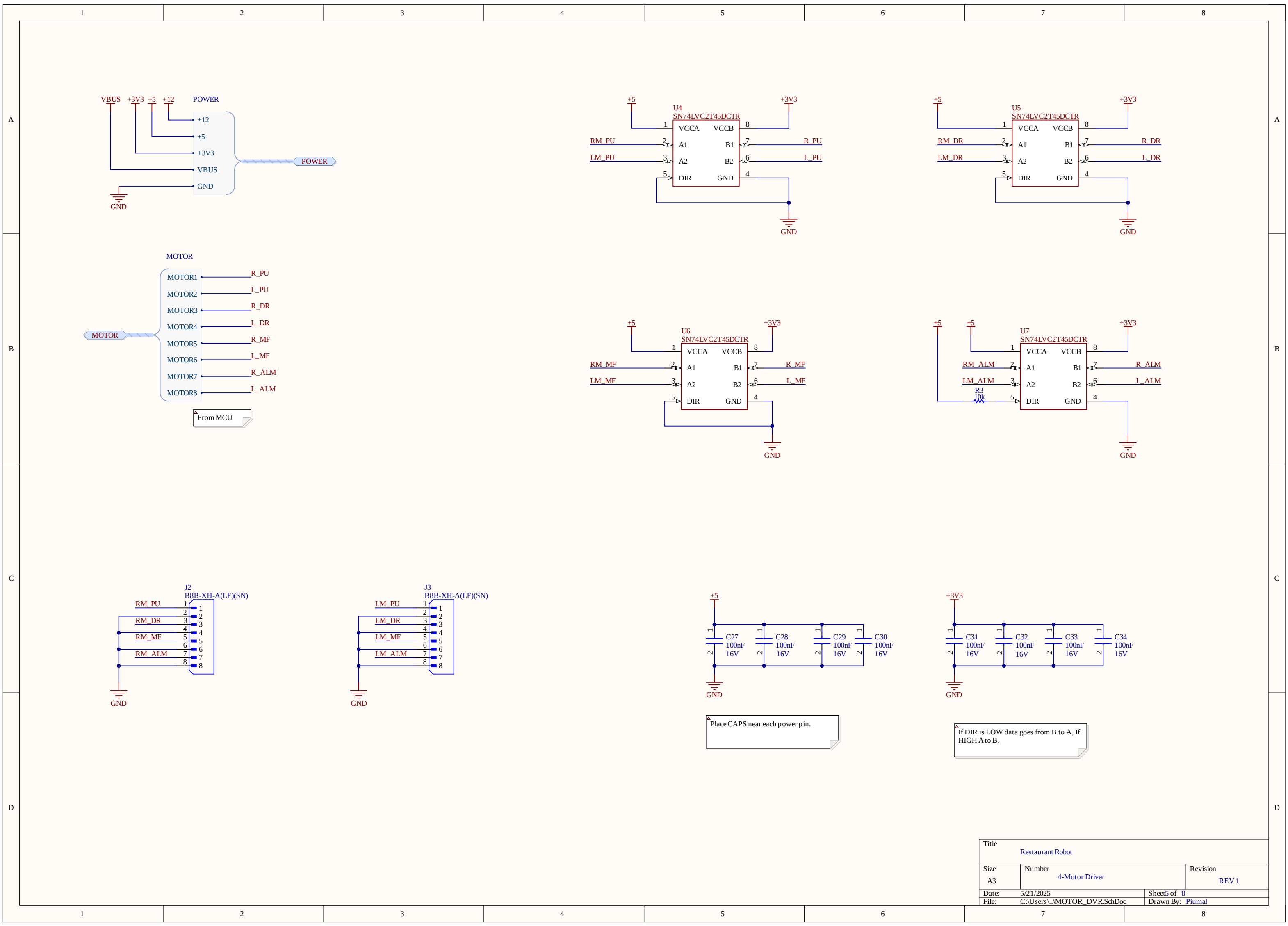
Need to add JTAG/SWD as alternating programming port. Read about ERASE pin and JTAGSEL. And what are need to flash firmwear into clean chip

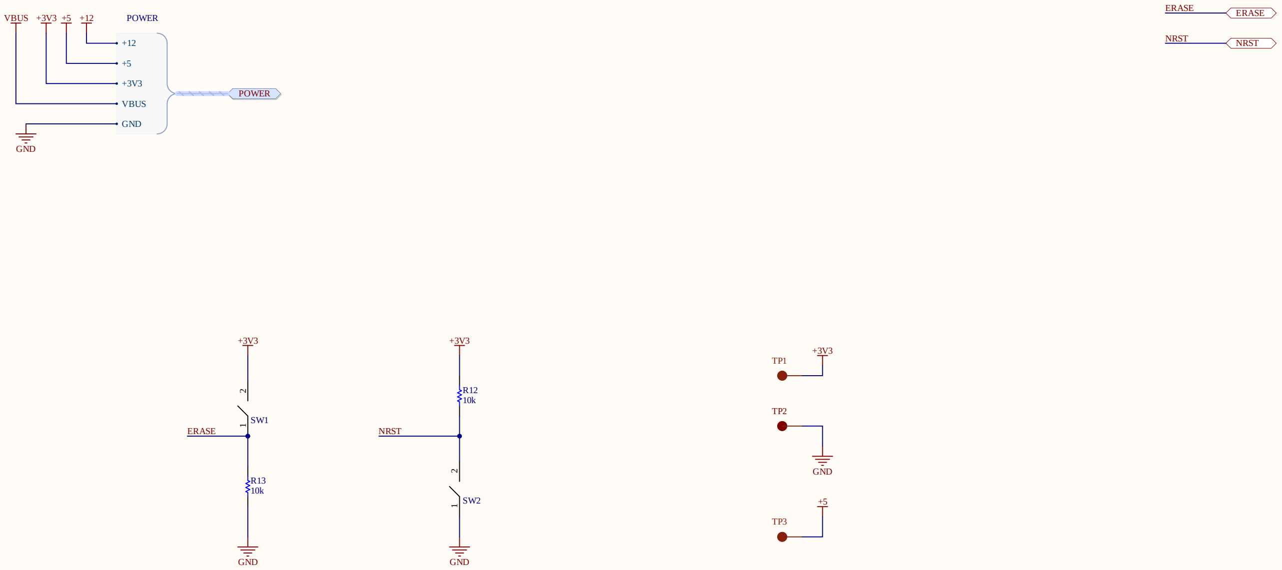
Title	Restaurant Robot		
Size	Number		Revision
A2	2-SAM3X_Main		REV 1
Date:	5/21/2025	Sheet2 of 8	
File:	C:\Users\...\SAM3X-MCU.SchDoc	Drawn By:	Piumal


POWER
+12
+5
+3V3
VBUS
GND
USB_P
USB_N
Route as Diff_Pair
NRST
U1B
ATSAM3X8EA-AU
TST
XIN
XOUT
SHDN
FWUP
XIN32
XOUT32
NRSTB
VBUS
DFSDP
DFSDM
DHSDP
DHSDM
JTAGSEL
ADVREF
VBG
R2
10k
C1
100nF
16V
Y1
12MHZ
C5
18 pF
50 V
C6
Y2
32.768KHZ
C3
6 pF
C4
U1C
+VDDOUT
VDDOUT
VDDCORE
VDDANA
VDDBU
VDDPLL
VDDUTMI
VDDIO
GNDANA
GNDBU
GNDPLL
GNDUTMI
C2
C7
2.2uF
C8
C9
C10
C11
C13
C14
C12
C18
C19
C20
C21
C17
C22
C23
Place these near each power pin of mcu
Title
Restaurant Robot
Size
A3
Number
2-SAM3X_Power and USB
Revision
REV 1
Date: 5/21/2025
Sheet3 of 8
File: C:\Users\..\SAM3X_MCU2.SchDoc
Drawn By: Piumal

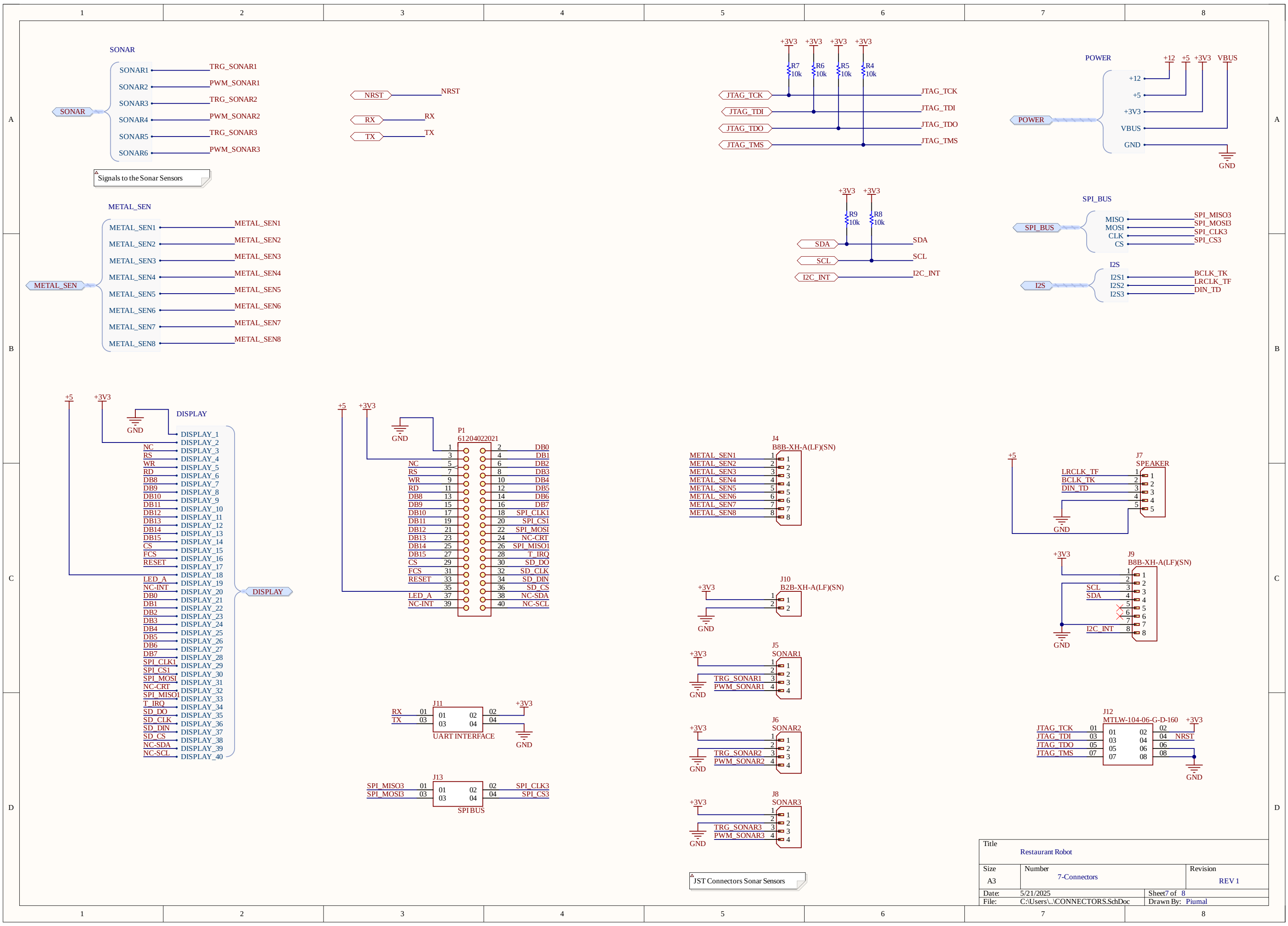


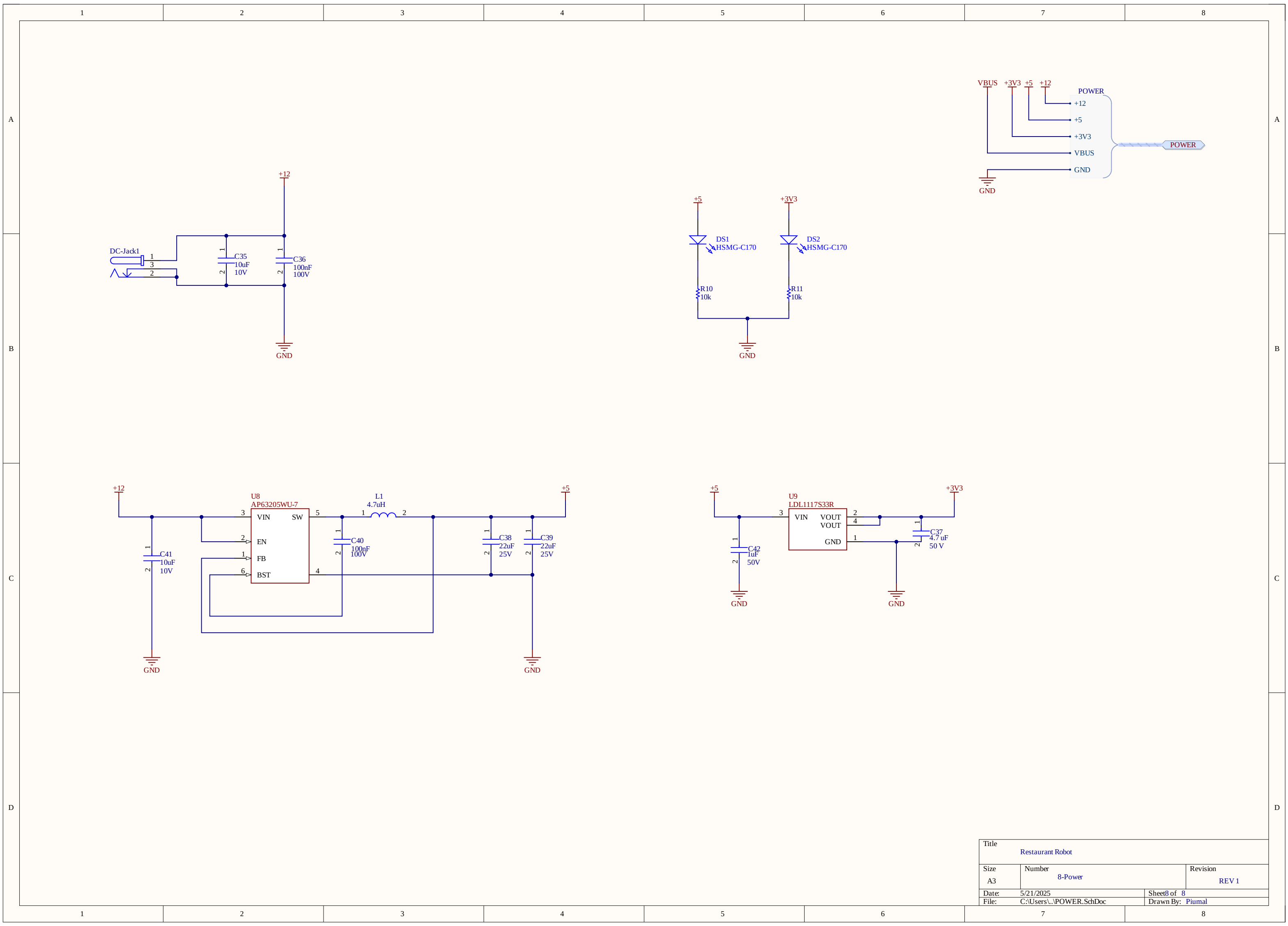
VBUS
+3V3
+5
+12
POWER
+12
+5
+3V3
VBUS
GND
GND
POWER
USB_P
USB_N
USB_P
USB_N
Route as Diff_Pair
FLASH
SPI_MISO2
SPI_MOSI2
SPI_CLK2
SPI_CS2
FLASH1
FLASH2
FLASH3
FLASH4
FLASH
VBUS
U2
LDL1117S33R
VIN
VOUT
VOUT
GND
+3V3
C16
1uF
50V
C15
4.7 uF
50 V
GND
GND
VBUS
RV1
CG0603MLC-05E
C24
10uF
10V
C25
100nF
16V
GND
J1
VBUS
D-
D+
ID
GND
SHLD1
SHLD2
SHLD3
SHLD4
SHLD5
SHLD6
105017-0001
XUSB_N
XUSB_P
GND
D1
I/O1_1
GND
I/O2_1
I/O1_2
VBUS
I/O2_2
USBLC6-2SC6Y
GND
VBUS
USB_N
USB_P
+3V3
U3
W25Q128FVSIG
SPI_MOSI2
SPI_MISO2
DI (IO0)
DO (IO1)
WP (IO2)
HOLD/RESET (IO3)
VCC
CS
CLK
GND
SPI_CS2
SPI_CLK2
C26
100nF
16V
GND
Title
Restaurant Robot
Size
A3
Number
4-USB
Revision
REV 1
Date: 5/21/2025
Sheet4 of 8
File: C:\Users\..\USB.SchDoc
Drawn By: Piumal


VBUS +3V3 +5 +12 POWER
+12
+5
+3V3
VBUS
GND
POWER
GND
MOTOR
MOTOR1 R_PU
MOTOR2 L_PU
MOTOR3 R_DR
MOTOR4 L_DR
MOTOR5 R_MF
MOTOR6 L_MF
MOTOR7 R_ALM
MOTOR8 L_ALM
From MCU
U4 SN74LVC2T45DCTR
U5 SN74LVC2T45DCTR
U6 SN74LVC2T45DCTR
U7 SN74LVC2T45DCTR
VCCA VCCB A1 B1 A2 B2 DIR GND
RM_PU LM_PU R_PU L_PU
RM_DR LM_DR R_DR L_DR
RM_MF LM_MF R_MF L_MF
RM_ALM LM_ALM R_ALM L_ALM
R3 10k
J2 B8B-XH-A(LF)(SN)
J3 B8B-XH-A(LF)(SN)
C27 100nF 16V
C28 100nF 16V
C29 100nF 16V
C30 100nF 16V
C31 100nF 16V
C32 100nF 16V
C33 100nF 16V
C34 100nF 16V
Place CAPS near each power pin.
If DIR is LOW data goes from B to A, If HIGH A to B.
Title Restaurant Robot
Size A3
Number 4-Motor Driver
Revision REV 1
Date: 5/21/2025
Sheet 5 of 8
File: C:\Users\..\MOTOR_DVR.SchDoc
Drawn By: Piumal

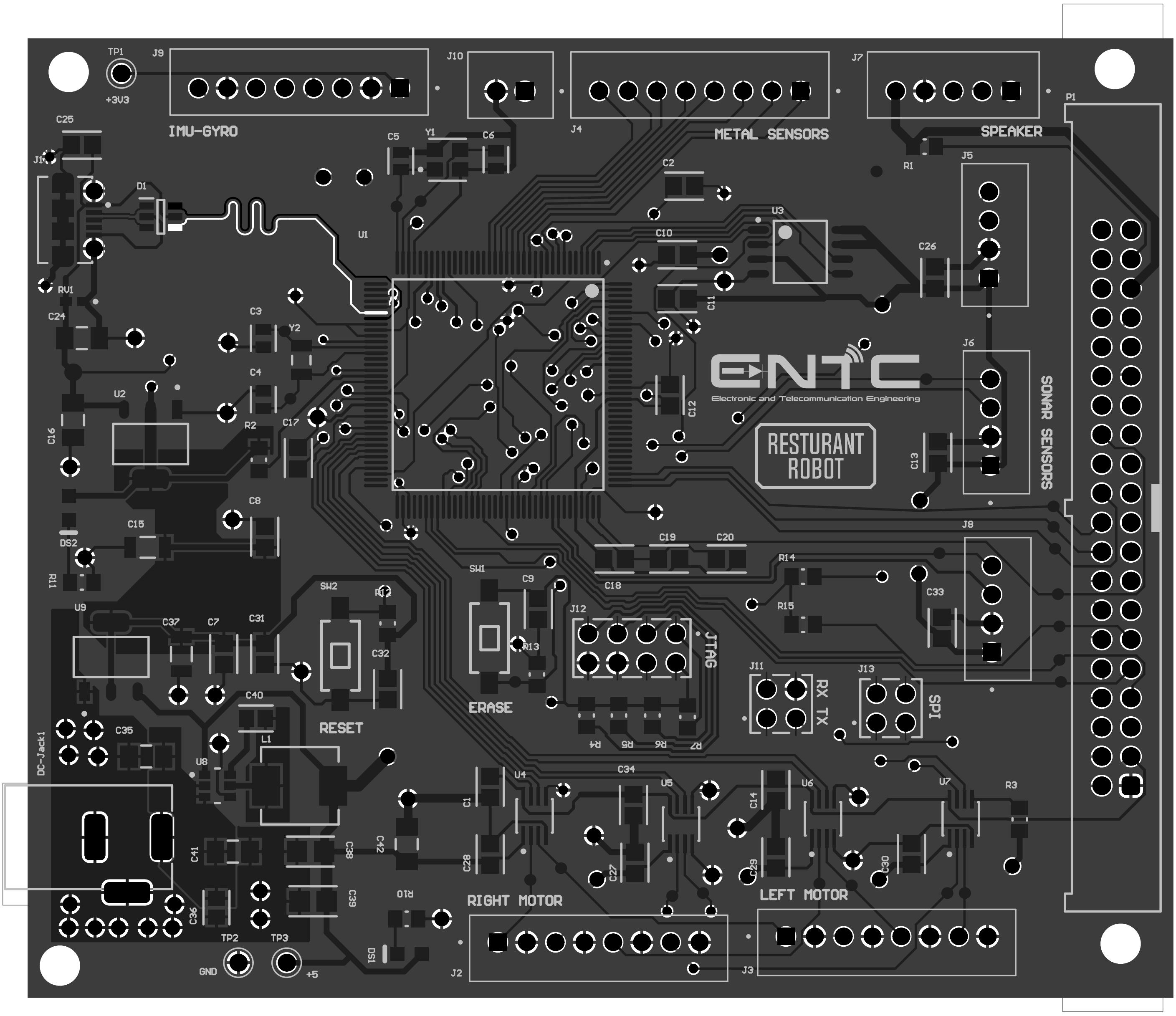


Title	Restaurant Robot	
Size A3	Number 6-Buttons and Test Points	Revision REV 1
Date: 5/21/2025		Sheet 6 of 8
File: C:\Users\..\BUTTONS.SchDoc		Drawn By: Piumal


SONAR
SONAR1 SONAR2 SONAR3 SONAR4 SONAR5 SONAR6
TRG_SONAR1 PWM_SONAR1 TRG_SONAR2 PWM_SONAR2 TRG_SONAR3 PWM_SONAR3
Signals to the Sonar Sensors
METAL_SEN
METAL_SEN1 METAL_SEN2 METAL_SEN3 METAL_SEN4 METAL_SEN5 METAL_SEN6 METAL_SEN7 METAL_SEN8
NRST
RX
TX
JTAG_TCK JTAG_TDI JTAG_TDO JTAG_TMS
R7 10k R6 10k R5 10k R4 10k
R9 10k R8 10k
SDA SCL I2C_INT
POWER
+12 +5 +3V3 VBUS GND
SPI_BUS
MISO MOSI CLK CS
SPI_MISO3 SPI_MOSI3 SPI_CLK3 SPI_CS3
I2S
I2S1 I2S2 I2S3
BCLK_TK LRCLK_TF DIN_TD
DISPLAY
P1 61204022021
J4 B8B-XH-A(LF)(SN)
J7 SPEAKER
J9 B8B-XH-A(LF)(SN)
J10 B2B-XH-A(LF)(SN)
J5 SONAR1
J6 SONAR2
J8 SONAR3
J11 UART INTERFACE
J13 SPI BUS
J12 MTLW-104-06-G-D-160
JST Connectors Sonar Sensors
Title Restaurant Robot
Size A3
Number 7-Connectors
Revision REV 1
Date: 5/21/2025
Sheet 7 of 8
File: C:\Users\..\CONNECTORS.SchDoc
Drawn By: Piumal



VBUS
+3V3
+5
+12
POWER
+12
+5
+3V3
VBUS
GND
GND
POWER
+12
DC-Jack1
C35
10uF
10V
C36
100nF
100V
GND
+5
+3V3
DS1
HSMG-C170
DS2
HSMG-C170
R10
10k
R11
10k
GND
+12
U8
AP63205WU-7
VIN
SW
EN
FB
BST
L1
4.7uH
C40
100nF
100V
C38
22uF
25V
C39
22uF
25V
+5
C41
10uF
10V
GND
GND
+5
U9
LDL1117S33R
VIN
VOUT
VOUT
GND
+3V3
C42
1uF
50V
C37
4.7 uF
50 V
GND
GND
Title
Restaurant Robot
Size
A3
Number
8-Power
Revision
REV 1
Date: 5/21/2025
Sheet8 of 8
File: C:\Users\..\POWER.SchDoc
Drawn By: Piumal



IMU-GYRO
METAL SENSORS
SPEAKER
ENTC
Electronic and Telecommunication Engineering
RESTURANT
ROBOT
SONAR SENSORS
JTAG
RX TX
SPI
RESET
ERASE
RIGHT MOTOR
LEFT MOTOR
+3V3
GND
+5